

Vector Differential Calculus

Vector Function of One Variable in Space.

Defn A Vector-valued function or vector function is a function whose domain is a set of real numbers and whose range is a set of vectors.

* A vector-valued fun in $\mathbb{R}^3$ is denoted by $r(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} = (f(t), g(t), h(t))$, where f, g, h are real valued functions & we call them component functions of r .

Eg. $r(t) = t^2\mathbf{i} + e^{-2t}\mathbf{j} + \ln(t+1)\mathbf{k}$ is a vector-valued function and the component functions of r are t^2, e^{-2t} & $\ln(t+1)$.

Defn (Domain of Vector-valued fun)
Domain of a vector valued fun $r(t)$ is the set of all values of t for which the expression $r(t)$ is defined.

Eg. Let $r(t) = \sqrt{t-1}\mathbf{i} + \ln(t+2)\mathbf{j} + 4t^2\mathbf{k}$
Then find domain of r .

Soln Domain of $r(t)$ is the set of all points in $\mathbb{R}$, where $\sqrt{t-1}, \ln(t+2)$ & $4t^2$ are defined.

$$D_r = [1, \infty) \cap [-2, \infty) \cap \mathbb{R} = [1, \infty)$$

Graph of Vector-valued function

Suppose $r(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ is a vector-valued function defined on interval I . Then the graph of $r(t)$ is

$C = \text{graph } r(t) = \{(f(t), g(t), h(t)); t \in I\}$
- is a curve in three dimensional space.

- is a curve in space which is traced out by $r(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$

That is, as t varies, the function $r(t)$ traces out a curve in space.

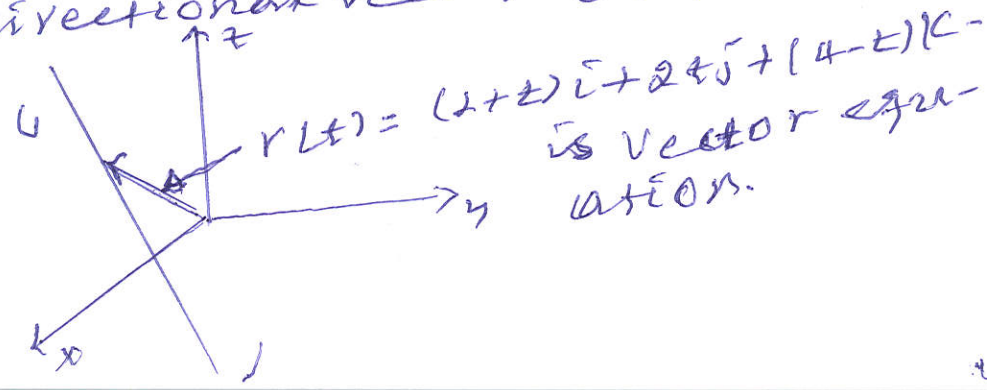
OR
C - is a curve in space described by a parametric equation:

$$\begin{aligned} x &= f(t) \\ y &= g(t) \\ z &= h(t), t \in I \end{aligned} \quad \text{is the parametric equation of C.}$$

And, $r(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ - is the vector equation of the same curve C.

Ex1. $r(t) = (1+t)\mathbf{i} + 2t\mathbf{j} + (4-t)\mathbf{k}$ - is a vector-valued function of one variable.

Soln The curve that is traced out by $r(t)$ is a line that passes through the pt $(1, 0, 4)$ and with directional vector $(1, 2, -1)$.



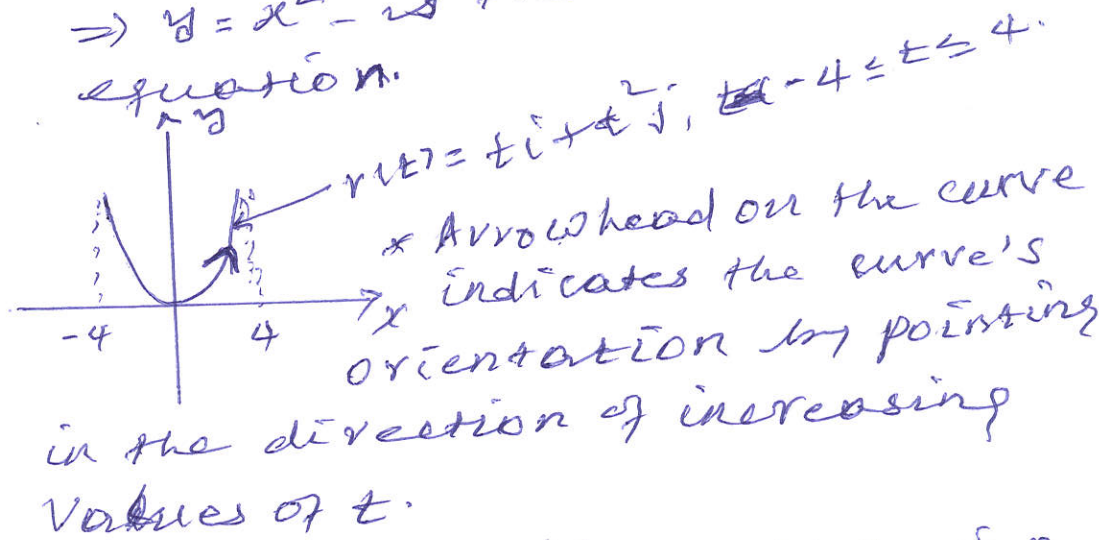
parametric equation:

$$\begin{aligned} x(t) &= 1+t = f(t) \\ y(t) &= 2t = g(t) \\ z(t) &= 4-t = h(t), t \in \mathbb{R} \end{aligned}$$

Ex2. Sketch the curve represented by the vector-valued function $r(t) = t\mathbf{i} + t^2\mathbf{j}, [-4, 4]$.

Soln Parametric equation:

$$\begin{aligned} x &= t, y = t^2 \\ \Rightarrow y &= x^2 \text{ - is the rectangular equation.} \end{aligned}$$



$\Rightarrow C: r(t) = t\mathbf{i} + t^2\mathbf{j}, -4 \leq t \leq 4$ - is a vector equation.

$$\begin{aligned} x(t) &= t \\ y(t) &= t^2 \end{aligned} \quad \text{is the parametric eqn.} \\ y = x^2 \quad \text{is the rectangular eqn.}$$

Parametric Representation of curve

1. Line segment

$$\overrightarrow{QP} = (x_1 - x_0, y_1 - y_0, z_1 - z_0)$$

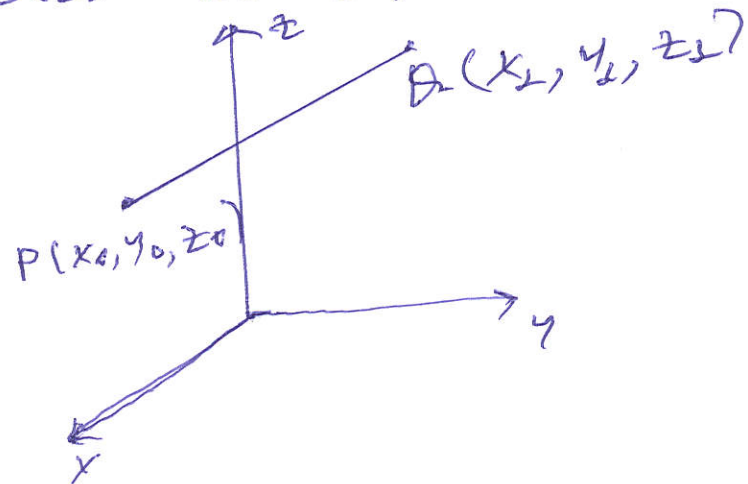
The line segment $\overline{PQ}$ is described by $r(t) = (x_0 + (x_1 - x_0)t)\mathbf{i} + (y_0 + (y_1 - y_0)t)\mathbf{j} + (z_0 + (z_1 - z_0)t)\mathbf{k}$, $0 \leq t \leq 1$, is vector equation.

Parametric equation:

$$x(t) = x_0 + (x_1 - x_0)t$$

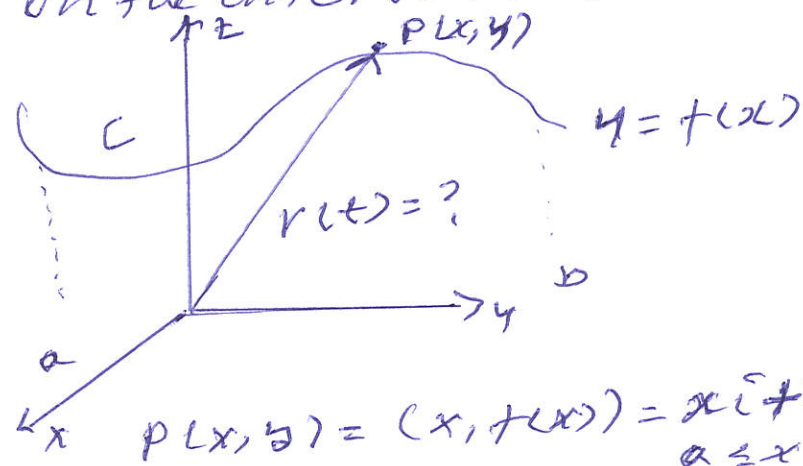
$$y(t) = y_0 + (y_1 - y_0)t$$

$$z(t) = z_0 + (z_1 - z_0)t, 0 \leq t \leq 1$$



2. Portion of graph of continuous function.

Suppose C is a portion of the graph of a continuous function $f(x)$ on the interval $[a, b]$.



$$P(x, y) = (x, f(x)) = x\mathbf{i} + f(x)\mathbf{j}, a \leq x \leq b.$$

Put $x = t$, $y = f(t)$, $a \leq t \leq b$.

Equation of C :

Vector equation: $r(t) = t\mathbf{i} + f(t)\mathbf{j}$, $a \leq t \leq b$.

Parametric equation:

$$x(t) = t$$

$$y(t) = f(t), a \leq t \leq b.$$

Q1. Find the parametric representation of the curve $C: y = x^3 + 5$, from $(0, 5)$ to $(2, 13)$

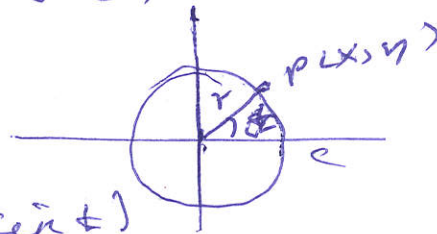
Soln Let $x = t$, $y = x^3 + 5 = t^3 + 5$. Then
 $r(t) = (x, y) = (t, t^3 + 5) = t\hat{i} + (t^3 + 5)\hat{j}$,
 $t \in [0, 2]$

$C: r(t) = t\hat{i} + (t^3 + 5)\hat{j}$, $0 \leq t \leq 2$

p. eqn: $x(t) = t$, $y(t) = t^3 + 5$, $0 \leq t \leq 2$

3. Circle: $x^2 + y^2 = r^2$

$x = r \cos t$, $y = r \sin t$,
 $0 \leq t \leq 2\pi$



$C: (x, y) = (r \cos t, r \sin t)$

$\Rightarrow r(t) = r \cos t \hat{i} + r \sin t \hat{j}$, $0 \leq t \leq 2\pi$

Q4: $x^2 + y^2 = 16$, $C: r(t) = 4 \cos t \hat{i} + 4 \sin t \hat{j}$,
 $0 \leq t \leq 2\pi$.

4. Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $C: r(t) = a \cos t \hat{i} + b \sin t \hat{j}$,
 $0 \leq t \leq 2\pi$.

Soln Let $u = \frac{x}{a}$, $v = \frac{y}{b} \Rightarrow u^2 + v^2 = 1$ is a unit circle.

$\& u(t) = \cos t$, $v(t) = \sin t$, $0 \leq t \leq 2\pi$.

$\Rightarrow \frac{x}{a} = u \Rightarrow x = a \cos t$, $\frac{y}{b} = v \Rightarrow y = b \sin t$

$C: r(t) = a \cos t \hat{i} + b \sin t \hat{j}$, $0 \leq t \leq 2\pi$.

Q5:

Defn Limit & Continuity

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Limit

If $r(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$ is a vector valued fun, then
 $\lim_{t \rightarrow a} r(t) = l = l_1\hat{i} + l_2\hat{j} + l_3\hat{k}$

$= \lim_{t \rightarrow a} f(t)\hat{i} + \lim_{t \rightarrow a} g(t)\hat{j} + \lim_{t \rightarrow a} h(t)\hat{k}$

provided f, g, h have limits as $t \rightarrow a$.

* A vector-valued fun $r(t)$ is continuous at the pt $t = a$ if

1. $\lim_{t \rightarrow a} r(t)$ exists $\&$

2. $\lim_{t \rightarrow a} r(t) = r(a)$

Q1. Let $r(t) = (4t + 5 \sin t)\hat{i} + e^{4t} \hat{j} + t^2 \hat{k}$.

Then find $\lim_{t \rightarrow \pi} r(t)$

Soln $\lim_{t \rightarrow \pi} r(t) = 4\pi \hat{i} + e^{4\pi} \hat{j} + \pi^2 \hat{k}$

Is $r(t)$ continuous?

$r(\pi) = 4\pi \hat{i} + e^{4\pi} \hat{j} + \pi^2 \hat{k} = \lim_{t \rightarrow \pi} r(t)$
 $\therefore r(t)$ is continuous at $t = \pi$.

Rules of Limits

Let $r_1(t)$ & $r_2(t)$ be vector-valued funs.
in 3-space & let c be a scalar. Then

$$1. \lim_{t \rightarrow a} [r_1(t) \pm r_2(t)] = \lim_{t \rightarrow a} r_1(t) \pm \lim_{t \rightarrow a} r_2(t)$$

$$2. \lim_{t \rightarrow a} [c r_1(t)] = c \lim_{t \rightarrow a} r_1(t)$$

$$3. \lim_{t \rightarrow a} [r_1(t) \times r_2(t)] = \lim_{t \rightarrow a} r_1(t) \times \lim_{t \rightarrow a} r_2(t)$$

$$4. \lim_{t \rightarrow a} [r_1(t) \cdot r_2(t)] = \lim_{t \rightarrow a} (r_1(t)) \cdot \lim_{t \rightarrow a} r_2(t)$$

Eq. let $r_1(t) = t\mathbf{i} + 5t\mathbf{j} + 4t\mathbf{k}$, $r_2(t) = 3t^2\mathbf{i} + te^{2t}\mathbf{j} + \sin(\pi/2 t)\mathbf{k}$.

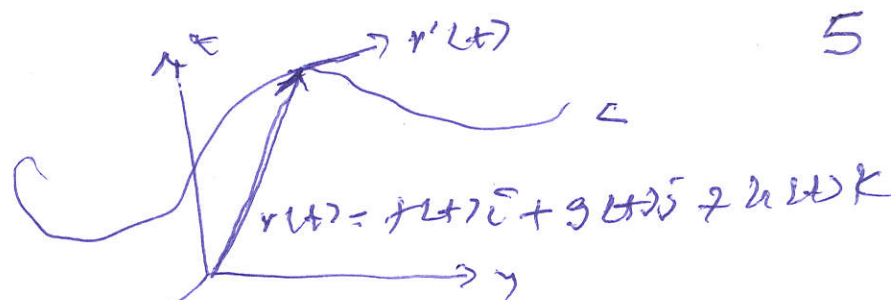
Then find $\lim_{t \rightarrow 2} [r_1 \times r_2]$.

Derivative of a vector function

If $r(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where f, g, h are differentiable funs, then

$$r'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}$$

$$= \lim_{h \rightarrow 0} \frac{r(t+h) - r(t)}{h} \text{ O.S.H.}$$



$r'(t) = \dots$ is a tangent vector to the curve C .

$\Rightarrow T(t) = \frac{r'(t)}{\|r'(t)\|}$ is called a unit tangent vector.

Unit normal: $N(t) = \frac{T'(t)}{\|T'(t)\|}$ is the direction in which particle changes direction suddenly (veering)
- always pts toward the concave side of C .

Curvature: $\kappa(t) = \frac{\|r'(t)\|}{\|r'(t)\|^3}$ the

rate of change of the tangent vector w.r.t the arc length.
(How many times near a moving object changes its direction with the given arc length).
- at what rate the tangent vector changes (twists)
- how curved a curve is

Eg. Let $\vec{r}(t) = 4t\vec{i} + 2\sin 2t\vec{j} - 2\cos 2t\vec{k}$.

Then find $\vec{i}, \vec{r}'(t), \vec{T}(t)$

Soln $\vec{i}, \vec{r}'(t) = 4\vec{i} + 4\cos 2t\vec{j} + 4\sin 2t\vec{k}$

- is a vector tangent to the curve C at time t .

(i) Unit tangent vector: $\vec{T}(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$
 $= \frac{4\vec{i} + 4\cos 2t\vec{j} + 4\sin 2t\vec{k}}{\sqrt{20}}$

Unit normal $\vec{N}(t) = \frac{\vec{T}'(t)}{\|\vec{T}'(t)\|}$

$\Rightarrow \vec{T}'(t) = \frac{0\vec{i} - 8\sin 2t\vec{j} + 8\cos 2t\vec{k}}{\sqrt{20}}$

$\& \|\vec{T}'(t)\| = \frac{8}{\sqrt{20}} [0\vec{i} - \sin 2t\vec{j} + \cos 2t\vec{k}]$

$\Rightarrow \vec{N}(t) = 0\vec{i} - \sin 2t\vec{j} + \cos 2t\vec{k}$.

Defn (Arc length)

The length s of the curve C which is given by the parametrization

$\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$ on $[a, b]$ is

defined by

$$s(t) = \int_a^b \sqrt{(\dot{x}(t))^2 + (\dot{y}(t))^2 + (\dot{z}(t))^2} dt = \int_a^b \|\vec{r}'(t)\| dt$$

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If we replace b (the fixed upper limit of integration) by a variable t , $a \leq t \leq b$, the integral becomes a function of t .

$s(t) = \int_a^t \|\vec{r}'(t)\| dt$ & is called the arc length function.

Eg. Let $C: \vec{r}(t) = a\cos t\vec{i} + a\sin t\vec{j} + ct\vec{k}$, $c \neq 0$, represent circular helix.

Then $\vec{r}'(t) = -a\sin t\vec{i} + a\cos t\vec{j} + c\vec{k}$ &

$\vec{r}'(t) \cdot \vec{r}'(t) = a^2 + c^2$. Hence the arc length of the circular helix is:

$$s(t) = \int_a^t \sqrt{\vec{r}'(t) \cdot \vec{r}'(t)} dt = \int_0^t \sqrt{a^2 + c^2} ds = t\sqrt{a^2 + c^2}$$

Defn (Curvature) $C: \vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$

$$\kappa(t) = \frac{\|\vec{T}'(t)\|}{\|\vec{r}'(t)\|^3} = \frac{\|\vec{r}'(t) \times \vec{r}''(t)\|}{\|\vec{r}'(t)\|^3}$$

- the rate of change of the tangent vector w.r. to the arc length.

(How many times per a moving obj. changes its direction with the given length)

- at what rate the tangent vector changes (twists). - how curved a curve is.

curvature of a line
 Ex 1. Let $r(t) = (x_0 + at)\mathbf{i} + (y_0 + bt)\mathbf{j} + (z_0 + ct)\mathbf{k}$
 Then find $\kappa(t)$.

$$r'(t) = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} \Rightarrow T(t) = \frac{r'(t)}{\|r'(t)\|} = \frac{1}{\sqrt{a^2 + b^2 + c^2}} (a\mathbf{i} + b\mathbf{j} + c\mathbf{k})$$

$$T'(t) = 0 \text{ for all } t \Rightarrow \text{hence } \kappa(t) = 0 \text{ for}$$

all t .

* This is clear from the fact that a particle moving on a straight line does not change its direction.

2. (curvatures of ellipse & circles)

$$r(t) = a\cos t\mathbf{i} + b\sin t\mathbf{j} + 0\mathbf{k}, \quad t \in \mathbb{R}.$$

$$\text{Then } r'(t) = -a\sin t\mathbf{i} + b\cos t\mathbf{j} \neq$$

$$\|r'(t)\| = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t}. \text{ Therefore}$$

$$T(t) = \frac{r'(t)}{\|r'(t)\|} = \frac{1}{\sqrt{a^2 \sin^2 t + b^2 \cos^2 t}} (-a\sin t\mathbf{i} + b\cos t\mathbf{j}) + 0\mathbf{k}$$

If $a = b$, then $T(t) = -\sin t\mathbf{i} + \cos t\mathbf{j} + 0\mathbf{k}$, then

$$T'(t) = -\cos t\mathbf{i} - \sin t\mathbf{j} + 0\mathbf{k} \neq \text{hence the curvature of the circle is } \kappa(t) = \frac{1}{a}.$$

Scalar and Vector fields

Scalar field, function

scalar field - is a ~~vector~~ ^{scalar} function of several variables / or simply a fun that assigns a ~~vector~~ ^{real no} to each pt in its domain (output of the fun is scalar). ($f: \mathbb{R} \subseteq \mathbb{R}^2 / \mathbb{R}^3 \rightarrow \mathbb{R}$)

Ex 1. $F(x, y) = x^2 + 2xy + y^3 - y + 3$
- is a scalar field fun.

2. $F(x, y, z) = x^2 + y^2 + z^2 + 2xyz + xy$
- is also a scalar field fun in $\mathbb{R}^3$.

Vector field function

- is a vector function of several variables / is a fun that assigns a vector to each pt in its domain.

Ex 1. $F(x, y, z) = f(x, y, z)\bar{i} + g(x, y, z)\bar{j} + h(x, y, z)\bar{k} = (f, g, h)$, where the scalar funs f, g & h are called the components of F . ($\bar{i}$ is the general form of vector-field fun in $\mathbb{R}^3$).

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Ex 1. Vector fields that are commonly occur in our physical world are:

* Electric field:

$$E(x, y, z) = \frac{q}{4\pi\epsilon_0 (x^2 + y^2 + z^2)^{3/2}} (x\bar{i} + y\bar{j} + z\bar{k})$$

* Gravitational field:

$$G(x, y, z) = -\frac{Gm}{(x^2 + y^2 + z^2)^{3/2}} (x\bar{i} + y\bar{j} + z\bar{k})$$

* Scalar fields:

$\bar{i}$, the fun $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $T(x, y) =$

Gradient of scalar field
Let $F(x, y, z)$ be a real valued function of three variables. Then the gradient of F , denoted by ∇F is a vector field defined by $\nabla F = \frac{\partial F}{\partial x}\bar{i} + \frac{\partial F}{\partial y}\bar{j} + \frac{\partial F}{\partial z}\bar{k} = (f_x, f_y, f_z)$

Ex 1. If $F(x, y, z) = 3x^2 - xy + yz^2$, then F is a scalar field & $\nabla F(x, y, z) = (6x - y)\bar{i} + (-x + z^2)\bar{j} + 2yz\bar{k}$ - is a vector field.

* The vector operator

$\nabla = \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k}$ - is called the del operator.

Divergence and curl of vector fields

Let $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a differentiable vector field given by

$$F(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$$

1. The divergence of F , denoted by $\text{div } F$, is the scalar field defined

$$\text{by } \text{div } F = \nabla \cdot F = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$$

2. The curl of F , denoted by $\text{curl } F$, is the vector field defined by

$$\text{curl } F = \nabla \times F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix}$$

$$= (h_y - g_z)\mathbf{i} - (h_x - f_z)\mathbf{j} + (g_x - f_y)\mathbf{k}$$

- measures the tendency of the vector field to rotate about an axis.

* $\text{div } F$ - measures the tendency of vector field to converge or diverge / unit volume in unit time.

* Remark:

* The vector field $F(x, y, z)$ is called rotation free if the curl is zero, $\text{curl } F = 0$.

* It is called incompressible if the divergence is zero, $\text{div } F = 0$.

Eg 1. The gravitational field = $F(x, y, z)$ is rotation free.

2. Magnetic field $\mathbf{B}$ - is always incompressible.

* Suppose the vector field F represents the velocity of a gas flowing in the xy -plane (x, y, z -space). Then it is

1. expanding if $\text{div } F > 0$

2. compressing if $\text{div } F < 0$

3. neither expanding nor compressing if $\text{div } F = 0$.

Conservative Vector fields & test for conservativeness

A vector field function F is conservative function if there exists a scalar field function f such that

$\nabla f = F$, & f is called a potential function for F .

Ex 1. Consider the vector field fun

$F(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$ & the scalar

field fun $f(x, y, z) = xyz$

now, $\nabla f = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$

$\Rightarrow F = \nabla f$, then F is conservative vector field & f is a potential function for F .

~~* Conservative fun has potential fun~~

Questions:

1. How do we know whether a given vector field F is conservative?
2. If so, how do we find a potential function f (so that $\nabla f = F$)?

Theorem (test for conservative field)

A vector field fun

$F(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$ is said to be

conservative iff $\text{curl } F = 0$. That is

$$P_y = N_z, M_z = P_x, \text{ \& } N_x = M_y$$

Ex 1. Consider the vector field

fun $F(x, y, z) = y^2z^3\mathbf{i} + 2xy^2z^3\mathbf{j} + 3xy^2z^2\mathbf{k}$

is F conservative?

$$\text{curl } F = (P_y - N_z)\mathbf{i} + (M_z - P_x)\mathbf{j} + (N_x - M_y)\mathbf{k}$$

$$= (6xy^2z^2 - 6xy^2z^2)\mathbf{i} + (3y^2z^2 - 3y^2z^2)\mathbf{j} + (2yz^3 - 2yz^3)\mathbf{k} = 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$$

$\Rightarrow \text{curl } F = 0$

$\therefore F(x, y, z) = y^2z^3\mathbf{i} + 2xy^2z^3\mathbf{j} + 3xy^2z^2\mathbf{k}$ is conservative.

ii, finding corresponding potential function f :

suppose $f(x, y, z)$ can be found, then
 $\nabla f = f_x \mathbf{i} + f_y \mathbf{j} + f_z \mathbf{k} = \mathbf{F} = y^2 z^3 \mathbf{i} + 2xy z^3 \mathbf{j} + 3xy^2 z^2 \mathbf{k}$

the two vectors equal:

$$f_x(x, y, z) = y^2 z^3 \quad \dots (1)$$

$$f_y(x, y, z) = 2xy z^3 \quad \dots (2)$$

$$f_z(x, y, z) = 3xy^2 z^2 \quad \dots (3)$$

From eqn (1), we get

$$\begin{aligned} f(x, y, z) &= \int y^2 z^3 dx + g(y, z) \\ &= y^2 z^3 x + g(y, z) \quad \dots (4) \end{aligned}$$

Now, differentiating (4) w.r. to y , we

have $f_y(x, y, z) = 2xy z^3 + g_y(y, z) \neq$
 comparison with (2) gives

$$g_y(y, z) = 0 \Rightarrow g(y, z) = \int 0 dy + K(z)$$

$$\Rightarrow g(y, z) = K(z)$$

And we rewrite eqn (4) as

$$f(x, y, z) = xy^2 z^3 + K(z).$$

Kindly, differentiating w.r. to z &
 comparing with eqn (3), we obtain

$$K'(z) = 0 \Rightarrow K(z) = C, \text{ where } C \text{ is arbitrary constant}$$

The desired fun $f(x, y, z)$ is: 11
 $f(x, y, z) = xy^2 z^3 + C.$

Ex2. Given $\mathbf{F}(x, y, z) = 2xy \mathbf{i} + (x^2 + 2yz) \mathbf{j} + y^2 \mathbf{k}.$

a, show that $\mathbf{F} = \nabla f$ conservative

b, find a fun f (corresponding potential fun) s. t $\mathbf{F} = \nabla f.$

Line Integrals

Defn 1 (smooth curve)

A curve C with parametrization $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ is called a smooth curve, if $r(t)$ is differentiable & $r'(t) = \frac{dr}{dt}$ is continuous & non-zero at every pt of C .

Defn 2. The line integral of a scalar function $f(x, y, z)$ over a curve C parametrized by $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, $a \leq t \leq b$ is defined

$$\text{by } \int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \|r'(t)\| dt$$

Defn 3. The line integral of a vector valued function $F(t) = f_1(t)\mathbf{i} + f_2(t)\mathbf{j} + f_3(t)\mathbf{k}$ over a curve C parametrized by $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, for $a \leq t \leq b$ is given by

$$\int_C \frac{F(r)}{\|r\|} \cdot dr = \int_a^b F(r(t)) \cdot \frac{dr}{dt} dt, \text{ where } \frac{dr}{dt} = (dx, dy, dz)$$

$$\Rightarrow \int_C (f_1 dx + f_2 dy + f_3 dz)$$

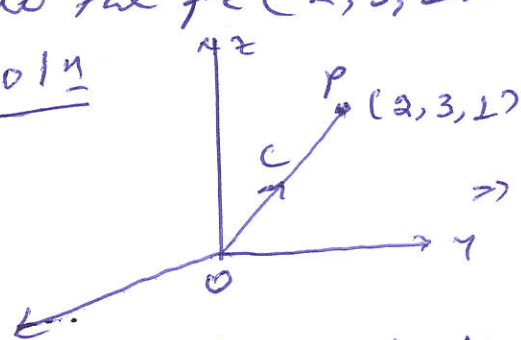
$$= \int_a^b (f_1 x'(t) + f_2 y'(t) + f_3 z'(t)) dt$$

* The curve C is called path of integration.

Ex 1. Evaluate $\int_C f(x, y, z) ds$, where

$f(x, y, z) = x - 3y^2 + z$ & C is the line segment joining the origin to the pt $(2, 3, 1)$.

Soln



$$C: r(t) = 0 + 0\mathbf{j} + 0\mathbf{k} + t(2\mathbf{i} + 3\mathbf{j} + \mathbf{k}), \quad 0 \leq t \leq 1$$

$$\Rightarrow C: r(t) = 2t\mathbf{i} + 3t\mathbf{j} + t\mathbf{k}, \quad 0 \leq t \leq 1.$$

$$\Rightarrow \|r'(t)\| = \|2\mathbf{i} + 3\mathbf{j} + \mathbf{k}\|$$

$$= \sqrt{4+9+1} = \sqrt{14} \neq 0, \text{ so, } C \text{ is a smooth curve.}$$

$$\Rightarrow \int_C f(x, y, z) ds = \int_0^1 f(2t, 3t, t) \sqrt{14} dt$$

$$= \sqrt{14} \int_0^1 (2t - 27t^2 + t) dt$$

$$= \sqrt{14} \int_0^1 (-27t^2 + 3t) dt$$

$$= \sqrt{14} \left[-9t^3 + \frac{3t^2}{2} \right]_0^1 = \sqrt{14} \left(-9 + \frac{3}{2} \right) = -\frac{15\sqrt{14}}{2}$$

Ex 2. Evaluate $\int_C F \cdot dr$, where

$$F(x, y, z) = 2y\mathbf{i} + 3x\mathbf{j} + 2z\mathbf{k} \text{ \& } C \text{ be}$$

$$\text{the helix } r(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + 3t\mathbf{k},$$

$$0 \leq t \leq 2\pi.$$

Soln $\int_C F \cdot dr = \int_0^{2\pi} F(r(t)) \cdot r'(t) dt$

$$= \int_0^{2\pi} (2\sin t\mathbf{i} + 3\cos t\mathbf{j} + 6t\mathbf{k}) \cdot (-\sin t\mathbf{i} + \cos t\mathbf{j} + 3\mathbf{k}) dt$$

$$= \int_0^{2\pi} (-2\sin^2 t + 3\cos^2 t + 18t) dt$$

$$= -2 \int_0^{2\pi} \sin^2 t dt + 3 \int_0^{2\pi} \cos^2 t dt + 18 \int_0^{2\pi} t dt$$

$$= -2 \left[\frac{1}{4} \sin 2t - \frac{1}{2} t \right]_0^{2\pi} + \frac{3}{4} \sin 2t \left[\frac{1}{4} + \frac{3}{2} t \right]_0^{2\pi} + 9t^2 \Big|_0^{2\pi}$$

$$= 2\pi + 3\pi + 36\pi^2 = \underline{\underline{36\pi^2 + 5\pi}}$$

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Ex 3. Evaluate $\int_L (xy^2 dx - \cos(xy) dy + y dz)$, where L is the line segment from $(0, 1, 1)$ to $(2, 2, -3)$.

Soln Parametric equation of L is given by $x = 2t, y = 1, z = 1 - 4t$, for $0 \leq t \leq 1$
 $dx = 2dt, dy = 0, dz = -4dt$. Then the line integral is

$$\int_L (xy^2 dx - \cos(xy) dy + y dz)$$

$$= \int_0^1 (2t(1-4t)2 - \cos(2t)(0) + 1(-4)) dt$$

$$= \int_0^1 (4t - 16t^2 - 4) dt = \left(2t^2 - \frac{16t^3}{3} - 4t \right) \Big|_0^1$$

$$= 2 - \frac{16}{3} - 4 = -\frac{22}{3}$$

Exercise: Evaluate

a) $\int_C (x^2 + 3y^2) ds$, where C is parametrized by $r(t) = \cos^3 t\mathbf{i} + \sin^3 t\mathbf{j}$, $0 \leq t \leq \frac{\pi}{2}$

b) $\int_C (4x^2 + y^3) dx - (x^3 + xy^2) dy + xy dz$, where C is parametrized by

$$r(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}, -\pi \leq t \leq \pi.$$

Line Integral of Vector field

A line integral of a vector field F over a smooth oriented curve C is denoted by $\int_C F \cdot dr$ & defined as

$$\begin{aligned} \int_C F \cdot dr &= \int_a^b F(x(t), y(t), z(t)) \cdot r'(t) dt \\ &= \int_a^b F(x(t), y(t), z(t)) \cdot (x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k}) dt \\ &= \int_a^b F(r(t)) \cdot r'(t) dt \end{aligned}$$

Ex 1. Let $F(x, y, z) = xy\mathbf{i} + yz\mathbf{j} + xz\mathbf{k}$ & $C: r(t) = t\mathbf{i} + t^2\mathbf{j} + 2t\mathbf{k}$, $0 \leq t \leq 1$. Then evaluate $\int_C F(x, y, z) \cdot dr$

Soln $F(x, y, z) = xy\mathbf{i} + yz\mathbf{j} + xz\mathbf{k}$

$$\Rightarrow F(x(t), y(t), z(t))$$

$$= t^3\mathbf{i} + 2t^3\mathbf{j} + 2t^4\mathbf{k}$$

$$\frac{dr}{dt} = r'(t) = (\mathbf{i} + 2t\mathbf{j} + 2\mathbf{k})$$

$$\Rightarrow dr = (\mathbf{i} + 2t\mathbf{j} + 2\mathbf{k}) dt$$

$$\Rightarrow \int_C F \cdot dr = \int_0^1 (t^3\mathbf{i} + 2t^3\mathbf{j} + 2t^4\mathbf{k}) \cdot (\mathbf{i} + 2t\mathbf{j} + 2\mathbf{k}) dt$$

$$= \int_0^1 (t^3 + 4t^4 + 4t^4) dt = \int_0^1 (t^3 + 8t^4) dt$$

$$= \left[\frac{t^4}{4} + \frac{8}{5}t^5 \right]_0^1 = \frac{5+32}{20} = \frac{37}{20}$$

42.

Evaluate $\int_C F \cdot dr$, where

$F(x, y, z) = x\mathbf{i} + 2z\mathbf{j} + y\mathbf{k}$, & C - is composed of the line segments from $(0, 0, 0)$ to $(0, 2, 0)$ & from $(0, -1, 0)$ to $(1, 1, 2)$.

Soln $C = C_1 \cup C_2$

$$C_1: r(t) = -t\mathbf{j}, 0 \leq t \leq 1$$

$$C_2: r(t) = t\mathbf{i} + (-1+2t)\mathbf{j} + 2t\mathbf{k}, 0 \leq t \leq 1$$

$$\int_C F \cdot dr = \int_{C_1} F \cdot dr + \int_{C_2} F \cdot dr$$

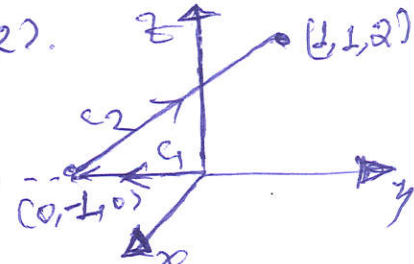
$$\begin{aligned} \neq \int_{C_1} F \cdot dr &= \int_0^1 (0\mathbf{i} + 0\mathbf{j} - t\mathbf{k}) \cdot (0\mathbf{i} - \mathbf{j} + 0\mathbf{k}) dt \\ &= \int_0^1 0 dt = 0 \end{aligned}$$

$$\neq \int_{C_2} F \cdot dr = \int_0^1 [(-t+2t^2)\mathbf{i} + 4t\mathbf{j} + (-1+2t)\mathbf{k}] \cdot (\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}) dt$$

$$\begin{aligned} &= \int_0^1 (-t+2t^2+8t-2+4t) dt = \int_0^1 (11t+2t^2-2) dt \\ &= \left[\frac{11t^2}{2} + \frac{2t^3}{3} - 2t \right]_0^1 = \frac{11}{2} + \frac{2}{3} - 2 = \frac{33+4-12}{6} \end{aligned}$$

$$= \frac{37-12}{6} = \frac{25}{6} //$$

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Theorem: If F is conservative that is

is $F = \nabla f$, for some fun f , then

- (i) $\int_C F \cdot dr$ - is independent of path
 (value of line integral depends only on initial & terminal pts)
- (ii) $\int_C F \cdot dr = 0$, for every closed, oriented curve C .

- (iii) $\int_C F \cdot dr = f(x_1, y_1, z_1) - f(x_0, y_0, z_0)$,
 where (x_0, y_0, z_0) is initial pt of C & (x_1, y_1, z_1) is terminal pt of C .

Ex 1. Let C be the curve from $(0, 3, 0)$ to $(1, 1, 2)$ & let $F(x, y, z) = 7x^2\mathbf{i} + xz^2\mathbf{j} + 2xyz\mathbf{k}$. Find $\int_C F \cdot dr$

Soln since $F = \nabla f$, where $f(x, y, z) = x^2yz^2$, then

$$\begin{aligned}\int_C F \cdot dr &= \int_C \nabla f \cdot dr = f(1, 1, 2) - f(0, 3, 0) \\ &= 4.\end{aligned}$$

Green's theorem (Transformation from line integral to double integral)

If R is a simple region in xy -plane bounded by a piecewise smooth curve C oriented counter clockwise & if $f(x, y)$ & $g(x, y)$ are continuous functions of x & y

($F = f\mathbf{i} + g\mathbf{j}$) having continuous partial derivatives in R , then

$$\int_C f dx + g dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

$$= \iint_R (g_x - f_y) dA$$

Ex. 1. Verify Green's Theorem for the vector field $F(x, y) = 2x\mathbf{i} - 3y\mathbf{j}$ & the region R bounded by the circle $C: \mathbf{r}(t) = a \cos t \mathbf{i} + a \sin t \mathbf{j}$, $0 \leq t \leq 2\pi$.

Soln $x = a \cos t$ & $y = a \sin t$.

$$F(\mathbf{r}(t)) = 2a \cos t \mathbf{i} - 3a \sin t \mathbf{j} \quad \& \quad dx = -a \sin t dt$$

$$dy = d(a \sin t) = a \cos t dt$$

$$\begin{aligned} \int_C F \cdot d\mathbf{r} &= \int_0^{2\pi} (-2a^2 \sin t \cos t - 3a^2 \sin t \cos t) dt \\ &= -5a^2 \int_0^{2\pi} \sin t \cos t dt = -5a^2 \left[\frac{\sin^2 t}{2} \right]_0^{2\pi} \end{aligned}$$

$$= 0.$$

$$f(x, y) = 2x, \quad g(x, y) = -3y$$

$$\Rightarrow g_x = 0 \quad \& \quad f_y = 0$$

$$\therefore \iint_R (g_x - f_y) dA = \iint_R 0 dA = 0$$

Surface in space

We can describe a surface in space in three different ways (forms):

Explicit form: $z = f(x, y)$

Implicit form: $f(x, y, z) = 0$

Parametric vector form:

$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$, for

$(u, v) \in R$, where R is the parameter domain & the variables u & v are the parametric variables.

Ex 1. ~~3.2~~ $S: z = x^2 + y^2$ is the paraboloid, which is the graph of function of ~~two~~ ^{two} variables $f(x, y) = x^2 + y^2$. Or, a level surface of fun of three variables $f(x, y, z) = x^2 + y^2 - z$.

Parametric vector form:

Let $x = u$, $y = v$. Then $z = x^2 + y^2 = u^2 + v^2$

$\Rightarrow S: \mathbf{r}(u, v) = u\mathbf{i} + v\mathbf{j} + (u^2 + v^2)\mathbf{k}$, for $u, v \in \mathbb{R}$.

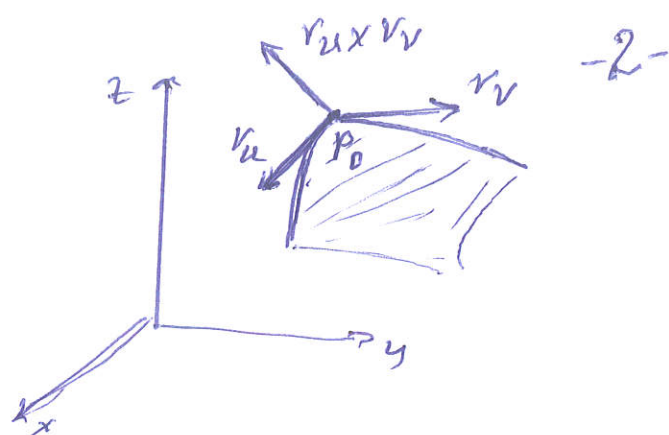
- gives the position of a point on the surface.

* Suppose S is the surface represented by an equation $S: f(x, y, z) = 0$. Then

$\nabla f(x, y, z) = f_x\mathbf{i} + f_y\mathbf{j} + f_z\mathbf{k}$ is perpendicular to (normal) to the surface S at the pt (x, y, z) .

Unit normal: $\hat{n} = \pm \frac{\nabla f(x, y, z)}{\|\nabla f(x, y, z)\|} = \pm \frac{f_x\mathbf{i} + f_y\mathbf{j} + f_z\mathbf{k}}{\sqrt{f_x^2 + f_y^2 + f_z^2}}$

$$\text{or } \hat{n} = \pm \frac{r_u \times r_v}{\|r_u \times r_v\|}$$



* If $\nabla f(x, y, z)$ is unique & continuous, then the surface is said to be orientable.

$\Rightarrow \hat{n} = \frac{\nabla f(x, y, z)}{\|\nabla f(x, y, z)\|}$ - is either directed outward from S or directed into S .

A surface in space can be oriented in two different ways:

1. Using normal vector directed outward

2. Using normal vector directed into S .

* A surface S , oriented by a normal vector directed outward has positive orientation. Otherwise, it ~~has~~ ^{has} negative orientation.

Surface Area

The surface area of the portion of the surface represented by $S: f(x, y, z) = 0$ over plane region

R is given by $S = \iint_R \sqrt{f_x^2 + f_y^2 + f_z^2} \, dA$, where

R is the projection of S into suitable xy -plane

$$\frac{dS}{dA} = \sqrt{f_x^2 + f_y^2 + f_z^2} \Rightarrow dS = \sqrt{f_x^2 + f_y^2 + f_z^2} \, dA$$

$$\Rightarrow \boxed{dS = \|\nabla f(x, y, z)\| \, dA}$$

$$= \|r_u \times r_v\| \, dA$$

* If S is a surface described by equation ~~$z = g(x, y)$~~

$z = g(x, y)$ (z -simple surface), then

$$S = \iint_R \sqrt{g_x^2 + g_y^2 + 1} dA \Rightarrow dS = \sqrt{g_x^2 + g_y^2 + 1} dA$$

where R - is the orthogonal projection of S into xy -plane.

* If $S: y = g(x, z)$, then $S = \iint_R \sqrt{g_x^2 + g_z^2 + 1} dA$

$$\Rightarrow dS = \sqrt{g_x^2 + g_z^2 + 1} dA$$

* If $S: x = g(y, z)$, then $S = \iint_R \sqrt{g_y^2 + g_z^2 + 1} dA$

$$\Rightarrow dS = \sqrt{g_y^2 + g_z^2 + 1} dA$$

Ex. Find the surface area of the portion of the surface $z = x^2 + y^2 = g(x, y)$ that lies below the plane $z = 4$.

Soln $S: z = x^2 + y^2 \Rightarrow S: f(x, y, z) = x^2 + y^2 - z$

$$\Rightarrow f_x = 2x, f_y = 2y, f_z = -1$$

$$\Rightarrow \sqrt{f_x^2 + f_y^2 + 1} = \sqrt{4x^2 + 4y^2 + 1}$$

$$S = \iint_R \sqrt{4x^2 + 4y^2 + 1} dA, \text{ where } R: x^2 + y^2 \leq 4$$

$$= \int_0^{2\pi} \int_0^2 \sqrt{4r^2 + 1} r dr d\theta$$

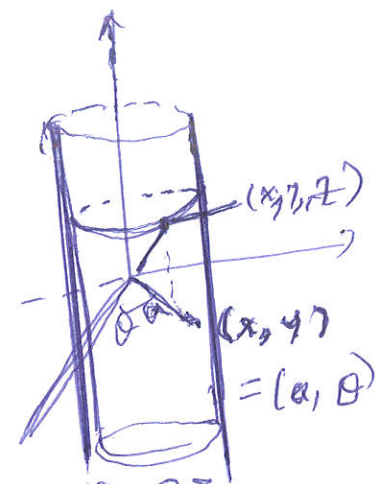
parametrization of some surfaces

1. The parametric representation of a circular cylinder:

$$S: x^2 + y^2 = a^2$$

Soln use cylindrical coordinates:
put $z = z, x = a \cos \theta, y = a \sin \theta$
 $0 \leq \theta \leq 2\pi, z \in \mathbb{R}$

$$r(\theta, z) = a \cos \theta \mathbf{i} + a \sin \theta \mathbf{j} + z \mathbf{k}$$



2. Cone: $z = \sqrt{x^2 + y^2}, 0 \leq z \leq a$

$$x = r \cos \theta, y = r \sin \theta \Rightarrow z = \sqrt{r^2} = r, 0 \leq \theta \leq 2\pi$$

$$\Rightarrow r(r, \theta) = r \cos \theta \mathbf{i} + r \sin \theta \mathbf{j} + r \mathbf{k}, 0 \leq r \leq a, 0 \leq \theta \leq 2\pi$$

3. Sphere: $x^2 + y^2 + z^2 = a^2$

Use spherical coordinates:

$$x = a \sin \phi \cos \theta, y = a \sin \phi \sin \theta, z = a \cos \phi$$

$$0 \leq \phi \leq \pi, 0 \leq \theta \leq 2\pi$$

$$r(\phi, \theta) = [a \sin \phi \cos \theta] \mathbf{i} + [a \sin \phi \sin \theta] \mathbf{j} + a \cos \phi \mathbf{k}, 0 \leq \phi \leq \pi, 0 \leq \theta \leq 2\pi$$

Ex 1. $x^2 + y^2 = 9, 0 \leq z \leq 4$

Soln $x = r \cos \theta = 3 \cos \theta, y = r \sin \theta = 3 \sin \theta$

$$\Rightarrow r(\theta, z) = 3 \cos \theta \mathbf{i} + 3 \sin \theta \mathbf{j} + z \mathbf{k}, 0 \leq \theta \leq 2\pi, 0 \leq z \leq 4$$

2. $x^2 + (y-2)^2 = 4, 0 \leq z \leq 2$

$$x^2 + y^2 - 4y + 4 = 4 \Rightarrow x^2 + y^2 - 4y = 0$$

$$\Rightarrow r^2 - 4r \sin \theta = 0 \Rightarrow r(r - 4 \sin \theta) = 0$$

$$\Rightarrow r = 0 \text{ or } r = 4 \sin \theta, 0 \leq \theta \leq \pi$$

$$\Rightarrow x = r \cos \theta = 4 \sin \theta \cos \theta = 2 \sin 2\theta$$

$$y = r \sin \theta = 4 \sin^2 \theta$$

$$z = z. \text{ Taking } u = \theta \text{ \& } v = z.$$

$$r(u, v) = 2 \sin 2u \mathbf{i} + 4 \sin^2 u \mathbf{j} + v \mathbf{k}, 0 \leq u \leq \pi, 0 \leq v \leq 2$$

4. Portion of plane: $\pi: ax+by+cz=d$

$$\Rightarrow \pi: z = \frac{a}{c}x + \frac{b}{c}y + \frac{d}{c} = g(u, v)$$

Set $x=u$, $y=v$, $z=g(u, v)$ & write the representation in parametric form:

$$r(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$$

$$= u\mathbf{i} + v\mathbf{j} + g(u, v)\mathbf{k}$$

$$= u\mathbf{i} + v\mathbf{j} + \left(\frac{a}{c}u + \frac{b}{c}v + \frac{d}{c}\right)\mathbf{k}, u, v \in \mathbb{R}.$$

ex. 1. $\pi: z - x - y + 3 = 0 \Rightarrow \pi: z = x + y - 3$

Let $u=x$, $v=y$, $z=u+v-3$. Then

$$r(u, v) = u\mathbf{i} + v\mathbf{j} + (u+v-3)\mathbf{k}$$

Surface Integrals of scalar field functions -6-

Any integral which is to be evaluated over a surface S is called surface integral. It is a natural generalization of a double integral over a plane region R .

Let $f(x, y, z)$ be a real valued fun which is defined & continuous on S , then the surface integral of $f(x, y, z)$ over S is denoted by $\iint_S f(x, y, z) ds \dots (1)$

To evaluate (1), we may reduce it to a double integral by introducing the vector representation of the surface S .

Evaluation of surface integral

If S is represented in parametric form by vector fun $r(u, v)$, then $ds = \|r_u \times r_v\| dA$

Hence, $\iint_S f(x, y, z) ds = \iint_R f(x(u, v), y(u, v), z(u, v)) \|r_u \times r_v\| dA$

where R is the region corresponding to S in the uv -plane or R is the region S expressed in the uv -plane or R is the projection of S into uv -plane

If a surface S is given by a representation $z = g(x, y)$, then we may set $x = u, y = v$ & write the representation in parametric form: $S: r(u, v) = ui + vj + g(u, v)k$

$$\Rightarrow r_u = i + g_u k, \quad r_v = j + g_v k \Rightarrow r_u \times r_v = \begin{vmatrix} i & j & k \\ 1 & 0 & g_u \\ 0 & 1 & g_v \end{vmatrix} = -g_u i - g_v j + k$$

$$\Rightarrow \|r_u \times r_v\| = \sqrt{g_u^2 + g_v^2 + 1}, \quad \text{if } u = x, v = y$$

$$= \sqrt{g_x^2 + g_y^2 + 1} \Rightarrow \boxed{ds = \sqrt{g_x^2 + g_y^2 + 1} dA}$$

$$\Rightarrow \iint_S f(x, y, z) ds = \iint_R f(x, y, g(x, y)) \sqrt{g_x^2 + g_y^2 + 1} dA, \text{ where } R \text{ is the orthogonal projection of } S \text{ into } xy\text{-plane.}$$

Similar argument shows that

If S be a surface with equation $y = g(x, z)$ & let R be its projection on the xz -plane, then

$$dS = \sqrt{g_x^2 + g_z^2 + 1} dA$$

$$\iint_S f(x, y, z) dS = \iint_R f(x, g(x, z), z) \sqrt{g_x^2 + g_z^2 + 1} dA$$

If S be a surface with equation $x = g(y, z)$ & let R be its projection on the yz -plane, then

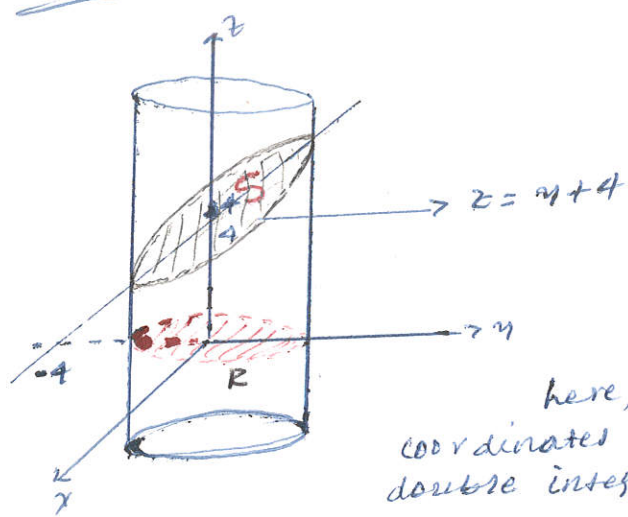
$$dS = \sqrt{g_y^2 + g_z^2 + 1} dA$$

$$\iint_S f(x, y, z) dS = \iint_R f(g(y, z), y, z) \sqrt{g_y^2 + g_z^2 + 1} dA$$

Surface Integral

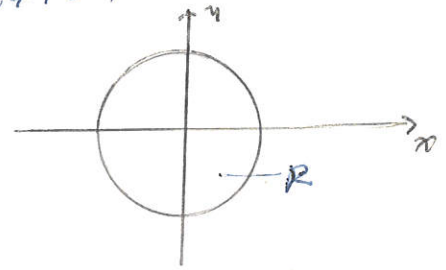
Q. 1. Evaluate $\iint_S (x^2 + y^2 + z^2) ds$, where S is the part of the plane $z = x + 4$ that is inside the cylinder $x^2 + y^2 = 4$

Soln let $g(x, y) = z = y + 4$, then $g_x = 0, g_y = 1 \Rightarrow ds = \sqrt{1+1+0} dxdy = \sqrt{2} dxdy$
i.e. S is z -simple surface



Thus $\iint_S (x^2 + y^2 + z^2) ds = \iint_R (x^2 + y^2 + z^2) \sqrt{2} dxdy$

where $R: x^2 + y^2 = 4$ in xy -plane



here, we use polar coordinates to evaluate the double integral

Description of R in polar coordinates

$0 \leq \theta \leq 2\pi, 0 \leq r \leq 2, \text{ and } dxdy = r dr d\theta$

$\Rightarrow \iint_S (x^2 + y^2 + z^2) ds = \iint_R (x^2 + y^2 + z^2) \sqrt{2} dxdy = \int_0^{2\pi} \int_0^2 (x^2 + y^2 + (y+4)^2) \sqrt{2} r dr d\theta$

$= \sqrt{2} \int_0^{2\pi} \int_0^2 (r^2 + r^2 \sin^2 \theta + 8r \sin \theta + 16) r dr d\theta$
 $= \sqrt{2} \int_0^{2\pi} \int_0^2 (r^3 (1 + \sin^2 \theta) + 8r^2 \sin \theta + 16r) dr d\theta$
 $= \sqrt{2} \int_0^{2\pi} \left[\frac{r^4}{4} (1 + \sin^2 \theta) + \frac{8}{3} r^3 \sin \theta + 8r^2 \right]_{r=0}^{r=2} d\theta$
 $= \sqrt{2} \int_0^{2\pi} \left[4(1 + \sin^2 \theta) + \frac{64}{3} \sin \theta + 32 \right] d\theta = 76\sqrt{2}\pi$ (please, complete it...)

Q. 2. Evaluate $\iint_S (x^2 - 2y + z) ds$, where S is the portion of the graph of $4x + y = 8$ bounded by the coordinate planes & the plane $z = 6$.

Soln The equation of the plane can be written as $y = 8 - 4x$ which is of the form $y = g(x, z)$, that is, S is y -simple surface, & $f(x, y, z) = x^2 - 2y + z$
Thus, $f_x = -4, f_y \text{ and } f_z = 0$

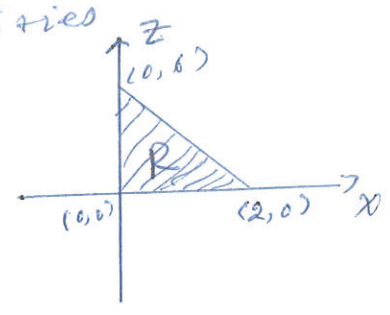
$$\Rightarrow ds = \sqrt{y^2 + z^2 + 1} dx dy = \sqrt{(1-y)^2 + 2^2 + 0} dx dy = \sqrt{5} dx dy$$

hence, $\iint_S (x^2 - 2y + z) ds = \iint_R (x^2 - 2(1-x) + 2) \sqrt{5} dx dy$, where

R is a plane region in xz -plane that described by the inequalities

$$0 \leq z \leq 6$$

$$0 \leq x \leq 2$$



$$\begin{aligned} \Rightarrow \iint_S (x^2 - 2y + z) ds &= \iint_R (x^2 + 8x - 16 + z) \sqrt{5} dx dz \\ &= \sqrt{5} \int_0^6 \int_0^2 (x^2 + 8x - 16 + z) dx dz \\ &= -44\sqrt{5} \end{aligned}$$

3. Evaluate $\iint_S (x^2 + y^2 + 1)^{1/2} ds$ where S is the portion of the paraboloid $2z = x^2 + y^2$ that lies inside the cylinder

$$x^2 + y^2 = 2y$$

soln write the equation for S in the form $z = \frac{x^2 + y^2}{2} = f(x, y)$ which is z -simple surface & then $f_x = x$ & $f_y = y$.

$$\Rightarrow ds = \sqrt{f_x^2 + f_y^2 + 1} dx dy = \sqrt{x^2 + y^2 + 1} dx dy$$

$$\text{Thus, } \iint_S (x^2 + y^2 + 1)^{1/2} ds = \iint_R (x^2 + y^2 + 1)^{1/2} \sqrt{x^2 + y^2 + 1} dx dy = \iint_R (x^2 + y^2 + 1) dx dy$$

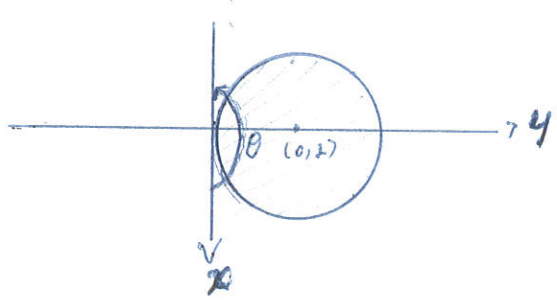
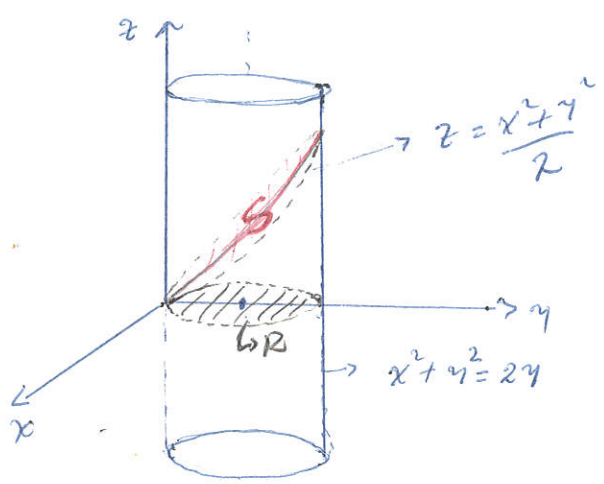
where $R: x^2 + y^2 = 2y$ is a circle of radius 1 centered at $(0, 1)$.

Let $x = r \cos \theta$, $y = r \sin \theta$, hence the eqn $x^2 + y^2 = 2y$ becomes
 $r^2 = 2r \sin \theta \Rightarrow r^2 - 2r \sin \theta = 0$
 $\Rightarrow r(r - 2 \sin \theta) = 0$
 $\Rightarrow r = 0$ or $r = 2 \sin \theta$
 $\Rightarrow 0 \leq r \leq 2 \sin \theta$

Description of R in polar C.

$$0 \leq r \leq 2 \sin \theta, \quad 0 \leq \theta \leq \pi$$

$$\begin{aligned} \iint_R (x^2 + y^2 + 1) dx dy &= \int_0^\pi \int_0^{2 \sin \theta} (r^2 + 1) r dr d\theta \\ &= \int_0^\pi \left(\frac{r^3}{3} + \frac{r^2}{2} \right) \Big|_0^{2 \sin \theta} d\theta \\ &= \int_0^\pi \left(\frac{16 \sin^4 \theta}{3} + \frac{4 \sin^2 \theta}{2} \right) d\theta \end{aligned}$$



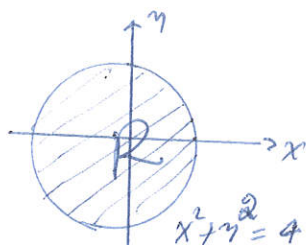
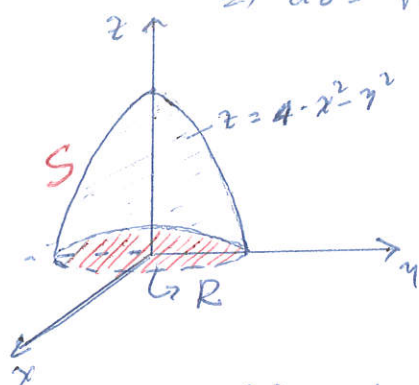
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4. Evaluate $\iint_S \frac{xy}{x^2+y^2} ds$ where S is a part of the paraboloid $z = 4 - x^2 - y^2$ that lies above xy -plane

Soln $S: z = 4 - x^2 - y^2 = f(x, y)$ & $f_x = -2x, f_y = -2y$

$\Rightarrow ds = \sqrt{f_x^2 + f_y^2 + 1} dx dy = \sqrt{4x^2 + 4y^2 + 1} dx dy$



Description of R in polar
 $0 \leq \theta \leq 2\pi, 0 \leq r \leq 2$
 $x = r \cos \theta, y = r \sin \theta$

Thus, $\iint_S \frac{xy}{x^2+y^2} ds = \iint_R \frac{xy}{x^2+y^2} \sqrt{4(x^2+y^2)+1} dx dy = \int_0^{2\pi} \int_0^2 r^2 \sin \theta \cos \theta \sqrt{4r^2+1} r dr d\theta$

$= \int_0^{2\pi} \int_0^2 r^2 \sqrt{4r^2+1} \sin \theta \cos \theta r dr d\theta$, let $u = 4r^2 + 1 \Rightarrow du = 8r dr$
 $\Rightarrow \frac{u-1}{4} = r^2 \Rightarrow \frac{du}{8} = r dr$

& $r=0, u=1$
 $r=2, u=4(2)^2+1=17$

$= \int_0^{2\pi} \int_1^{17} \left(\frac{u-1}{4}\right) \frac{1}{8} u \sin \theta \cos \theta \frac{du}{8} d\theta$

$= \frac{1}{32} \int_0^{2\pi} \int_1^{17} (u^{3/2} - u^{1/2}) \sin \theta \cos \theta du d\theta = \frac{1}{32} \int_0^{2\pi} \left(\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right) \Big|_1^{17} \sin \theta \cos \theta d\theta$

$= \frac{1}{32} \int_0^{2\pi} \left(\frac{2}{5} (4r^2+1)^{5/2} - \frac{2}{3} (4r^2+1)^{3/2} \right) \Big|_0^2 \sin \theta \cos \theta d\theta$

$= \frac{1}{32} \int_0^{2\pi} \left(\frac{2}{5} (17)^{5/2} - \frac{2}{3} (17)^{3/2} \right) \sin \theta \cos \theta d\theta = \frac{1}{32} \left(\frac{2}{5} (17)^{5/2} - \frac{2}{3} (17)^{3/2} \right) \int_0^{2\pi} \sin \theta \cos \theta d\theta$

$= \frac{1}{32} \left(\frac{2}{5} (17)^{5/2} - \frac{2}{3} (17)^{3/2} \right) \int_0^{2\pi} u du$

$= \frac{1}{32} \left(\frac{2}{5} (17)^{5/2} - \frac{2}{3} (17)^{3/2} \right) \left(\frac{\sin^2 \theta}{2} \right) \Big|_0^{2\pi} = \underline{\underline{0}}$

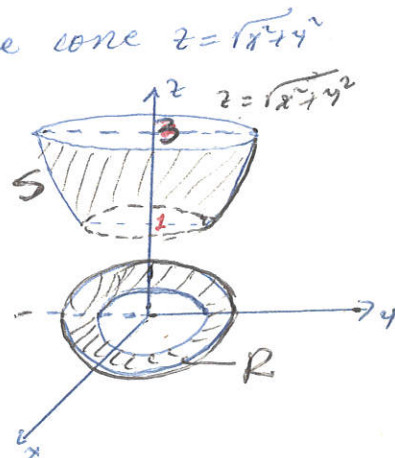
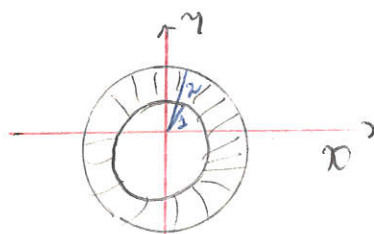
let $u = \sin \theta, du = \cos \theta d\theta$

5. evaluate $\iint_S z ds$ where S is the part of the cone $z = \sqrt{x^2 + y^2}$ b/n the planes $z=1$ & $z=3$.

Soln $S: z = \sqrt{x^2 + y^2} = f(x, y)$ & $f_x = \frac{x}{\sqrt{x^2+y^2}}, f_y = \frac{y}{\sqrt{x^2+y^2}}$

$\Rightarrow ds = \sqrt{f_x^2 + f_y^2 + 1} dx dy$

$= \sqrt{\frac{x^2}{x^2+y^2} + \frac{y^2}{x^2+y^2} + 1} dx dy = \sqrt{2} dx dy$



Description of R - in polar Coord.

$$0 \leq \theta \leq 2\pi$$

$$1 \leq r \leq 3 \quad \rightarrow \quad dxdy = r dr d\theta$$

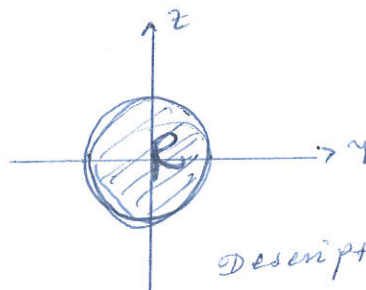
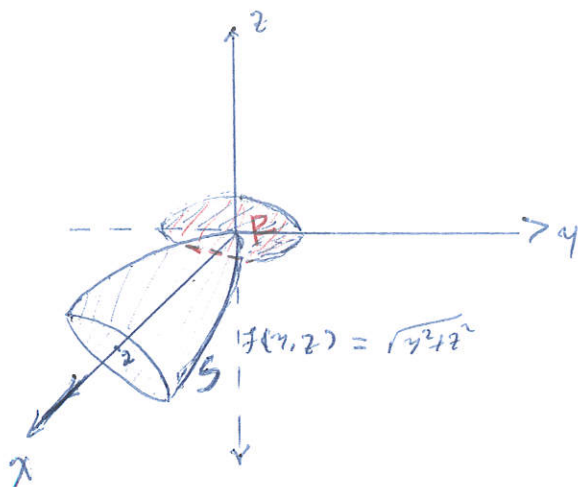
$$\begin{aligned} \text{Thus } \iint_S z \, ds &= \iint_R z \, r \, dr d\theta = \sqrt{2} \iint_R z \, dr d\theta = \sqrt{2} \int_0^{2\pi} \int_1^3 z \, dr d\theta = \sqrt{2} \int_0^{2\pi} \int_1^3 \sqrt{r^2 + z^2} \, dr d\theta \\ &= \sqrt{2} \int_0^{2\pi} \int_1^3 r \, dr d\theta = \sqrt{2} \int_0^{2\pi} \left[\frac{r^2}{2} \right]_1^3 d\theta = \sqrt{2} \int_0^{2\pi} \frac{8}{2} d\theta \\ &= \sqrt{2} \int_0^{2\pi} 4 \, d\theta = \sqrt{2} \cdot 4 \cdot 2\pi = 8\sqrt{2} \pi \end{aligned}$$

5. Evaluate $\iint_S x \, ds$, where S is the portion of the cone $x = \sqrt{y^2 + z^2}$

between the planes $x=0$, & $x=2$

Soln $S: x = \sqrt{y^2 + z^2} = f(y, z)$ - is x -simple surface & $f_y = \frac{-y}{\sqrt{y^2 + z^2}}$, & $f_z = \frac{-z}{\sqrt{y^2 + z^2}}$

$$\Rightarrow ds = \sqrt{f_y^2 + f_z^2 + 1} \, dy \, dz = \sqrt{\frac{y^2}{y^2 + z^2} + \frac{z^2}{y^2 + z^2} + 1} \, dy \, dz = \sqrt{2} \, dy \, dz$$



$$R: 2 = \sqrt{y^2 + z^2} \\ 4 = y^2 + z^2$$

Description of D - in polar C.

$$0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq 2$$

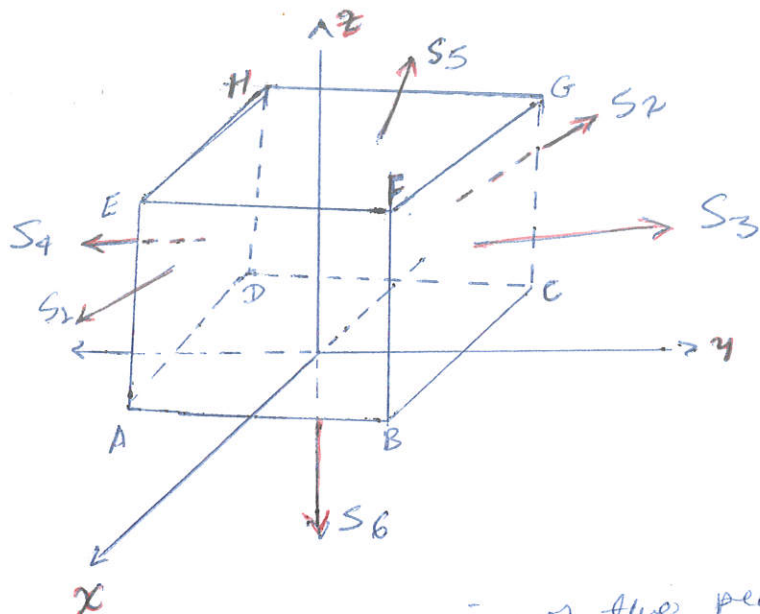
$$dxdy = r \, dr \, d\theta$$

$$\begin{aligned} \text{Thus } \iint_S x \, ds &= \iint_R x \, r \, dr d\theta = \iint_R \sqrt{y^2 + z^2} \, r \, dr d\theta = \sqrt{2} \iint_R r \, dr d\theta = \sqrt{2} \int_0^{2\pi} \int_0^2 r^2 \, dr d\theta \\ &= \sqrt{2} \int_0^{2\pi} \left[\frac{r^3}{3} \right]_0^2 d\theta = \sqrt{2} \cdot \frac{8}{3} \int_0^{2\pi} d\theta = \frac{16\sqrt{2} \pi}{3} \end{aligned}$$

6. Evaluate $\iint_S (x+y+z) \, ds$, where S is the cube centered at the origin

Soln S is composed of six surfaces, that is $S = S_1 + S_2 + S_3 + S_4 + S_5 + S_6$

$$\Rightarrow \iint_S (x+y+z) \, ds = \iint_{S_1} (x+y+z) \, ds + \iint_{S_2} (x+y+z) \, ds + \dots + \iint_{S_6} (x+y+z) \, ds$$



S_1 (DABFE): is a portion of the plane containing the pt $(1, 0, 0)$ & // to zy -plane or S_1 : is a set of pts $(1, y, z)$ such that $-1 \leq y \leq 1, -1 \leq z \leq 1$
Equation of the plane (or S_1): $N = \vec{i} + 0\vec{j} + 0\vec{k}$ - is normal vector
 $P_0 = (1, 0, 0)$

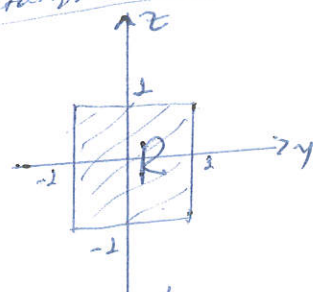
$$\Rightarrow S_1: (x, y, z) \cdot N = N \cdot P_0$$

$$\Rightarrow x + 0y + 0z = 1 \Rightarrow x = 1 \quad \text{i.e. } S_1: x = f(y, z) = 1 \text{ - is } x\text{-simple surface}$$

$$\Rightarrow f_y = 0, f_z = 0 \Rightarrow ds = \sqrt{0^2 + 0^2 + 1} dy dz = dy dz$$

$$\Rightarrow ds = dy dz$$

Description of R - in rectangular c. sy
 $-1 \leq y \leq 1, -1 \leq z \leq 1$



$$\begin{aligned} \iint_{S_1} (x+y+z) ds &= \iint_R (1+y+z) dy dz \\ &= \int_{-1}^1 \int_{-1}^1 (1+y+z) dy dz = \int_{-1}^1 \left(y + \frac{y^2}{2} + yz \right) \Big|_{-1}^1 dz \\ &= \int_{-1}^1 \left((1 + \frac{1}{2} + z) - (-1 + \frac{1}{2} - z) \right) dz = \int_{-1}^1 (2 + 2z) dz = (2z + z^2) \Big|_{-1}^1 \\ &= 2 + 1 - (-2 + 1) = 4 // \end{aligned}$$

S_2 (DDCGH): is a portion of the plane containing the pt $(-1, 0, 0)$ & // to zy -plane. Or S_2 : is a set of pts $(-1, y, z)$ such that $-1 \leq y \leq 1, -1 \leq z \leq 1$

$$\text{Equation of the plane (or } S_2): N = -\vec{i} + 0\vec{j} + 0\vec{k}$$

$$P_0 = (-1, 0, 0)$$

$$\Rightarrow S_2: (x, y, z) \cdot N = N \cdot P_0$$

$$\Rightarrow -x + 0y + 0z = -1$$

$$\Rightarrow S_2: x = -1 \Rightarrow S_2: x = f(y, z) = -1 \text{ is } x\text{-simple surface}$$

$$\Rightarrow f_y = 0, f_z = 0 \Rightarrow ds = \sqrt{1^2 + 0^2 + 0^2} dy dz = dy dz$$

$$\Rightarrow ds = dy dz$$

$$\begin{aligned} \iint_{S_2} (x+y+z) ds &= \iint_R (-1+y+z) dy dz = \int_{-1}^1 \int_{-1}^1 (-1+y+z) dy dz \\ &= \int_{-1}^1 \left(-y + \frac{y^2}{2} + yz \right) \Big|_{-1}^1 dz = \int_{-1}^1 \left((-1 + \frac{1}{2} + z) - (1 + \frac{1}{2} - z) \right) dz = \int_{-1}^1 (-2 + 2z) dz \\ &= (-2z + z^2) \Big|_{-1}^1 = -2 + 1 - (-2 + 1) = -4 \end{aligned}$$

S_3 ($\square BCGF$): is a portion of the plane containing the pt $(0, 1, 0)$ & // to xz -plane or S_3 : is a set of pts $(x, 1, z)$ such that

Equation of the plane (S_3) $-1 \leq x \leq 1, -1 \leq z \leq 1$.

$N = 0\mathbf{i} + \mathbf{j} + 0\mathbf{k}$ - is normal vector

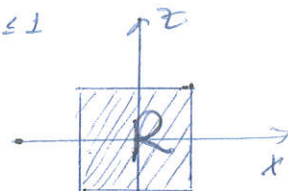
$P_0 = (0, 1, 0)$

$\Rightarrow S_3: (x, y, z) \cdot N = N \cdot P$

$\therefore 0x + y + 0z = 1 \Rightarrow y = 1$ i.e. $S_3: y = f(x, z) = 1$ - is y -simple surface

Description of R - in rect. C.S.

$-1 \leq x \leq 1, -1 \leq z \leq 1$



$\Rightarrow f_x = 0, f_z = 0 \Rightarrow ds = \sqrt{0^2 + 0^2 + 1} dx dz$

$$ds = dx dz$$

$$\begin{aligned} \iint_{S_3} (x+y+z) ds &= \iint_R (x+1+z) dx dz = \int_{-1}^1 \int_{-1}^1 (x+1+z) dx dz = \int_{-1}^1 \left(\frac{x^2}{2} + x + xz \right) \Big|_{-1}^1 dz \\ &= \int_{-1}^1 \left(\left(\frac{1}{2} + 1 + z \right) - \left(\frac{1}{2} - 1 - z \right) \right) dz = \int_{-1}^1 (2 + 2z) dz = (2z + z^2) \Big|_{-1}^1 \\ &= (2+1) - (-2+1) = 4 \end{aligned}$$

S_4 ($\square ADHE$): is a portion of the plane containing the pt $(0, -1, 0)$ & // to xz -plane or S_4 : is a set of pts $(x, -1, z)$ such that $-1 \leq x \leq 1, -1 \leq z \leq 1$.

Equation of the plane (or S_4):

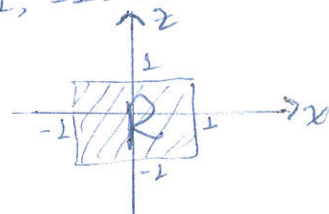
$N = 0\mathbf{i} - \mathbf{j} + 0\mathbf{k} \Rightarrow S_4: (x, y, z) \cdot N = N \cdot P$

$P_0 = (0, -1, 0) \therefore 0x - y + 0z = -1$

$\Rightarrow y = -1 \Rightarrow S_4: y = f(x, z) = -1$

Description of R - in R.C.S.

$-1 \leq x \leq 1, -1 \leq z \leq 1$



$\Rightarrow f_x = 0, f_z = 0 \Rightarrow ds = \sqrt{0^2 + 0^2 + 1} dx dz = dx dz$

$$\begin{aligned} \iint_{S_4} (x+y+z) ds &= \iint_R (x-1+z) dx dz = \int_{-1}^1 \int_{-1}^1 (x-1+z) dx dz = \int_{-1}^1 \left(\frac{x^2}{2} - x + xz \right) \Big|_{-1}^1 dz \\ &= \int_{-1}^1 \left(\left(\frac{1}{2} - 1 + z \right) - \left(\frac{1}{2} + 1 - z \right) \right) dz = \int_{-1}^1 (-2 + 2z) dz = (-2z + z^2) \Big|_{-1}^1 = -4 \end{aligned}$$

S_5 ($\square EFGH$): is a portion of the plane containing the pt $(0, 0, 1)$ & // xy -plane or S_5 : is the set of pts $(x, y, 1)$ such that $-1 \leq x \leq 1, -1 \leq y \leq 1$.

Equation of the plane (or S_5):

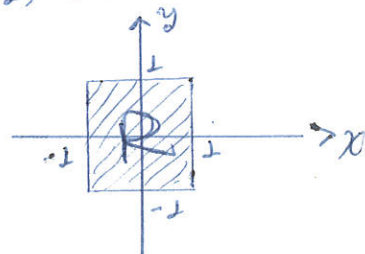
$N = 0\mathbf{i} + 0\mathbf{j} + \mathbf{k} \Rightarrow S_5: (x, y, z) \cdot N = N \cdot P$

$P_0 = (0, 0, 1) \therefore 0x + 0y + z = 1$

$\Rightarrow S_5: z = f(x, y) = 1$

Description of R -

$-1 \leq x \leq 1, -1 \leq y \leq 1$



$\Rightarrow f_x = 0 = f_y \Rightarrow ds = \sqrt{0^2 + 0^2 + 1} dx dy = dx dy$

$$ds = dx dy$$

$$\begin{aligned} \text{Thus } \iint_{S_5} (x+y+z) dS &= \iint_R (x+y+1) dx dy = \int_{-1}^1 \int_{-1}^1 (x+y+1) dx dy \\ &= \int_{-1}^1 (x^2/2 + xy + x) \Big|_{-1}^1 dy = \int_{-1}^1 ((1/2 + y + 1) - (1/2 - y - 1)) dy = \int_{-1}^1 (2y + 2) dy \\ &= (y^2 + 2y) \Big|_{-1}^1 = (1+2) - (1-2) = \underline{4} \end{aligned}$$

S_6 (□ABCD): is the portion of the plane containing the pt $(0, 0, -1)$ & // to the plane xy -plane or S_6 : is the set of pts $(x, y, -1)$ such that $-1 \leq x \leq 1, -1 \leq y \leq 1$.

Equation of the plane (or S_6)

$N = 0i + 0j - k$ - is normal vector

$P_0 = (0, 0, -1)$

$$\Rightarrow S_6: (x, y, z) \cdot N = N \cdot P_0$$

$$\therefore 0x + 0y - z = -1 \Rightarrow z = -1 \quad \text{i.e. } S_6: z = f(x, y) = -1$$

Description of R

$$-1 \leq x \leq 1, -1 \leq y \leq 1$$

$$\begin{aligned} \Rightarrow r_x = 0 = r_y &\Rightarrow ds = \sqrt{0^2 + 0^2} dx dy \\ &\Rightarrow \boxed{ds = dx dy} \end{aligned}$$

$$\begin{aligned} \text{Thus } \iint_{S_6} (x+y+z) dS &= \iint_R (x+y-1) dx dy = \int_{-1}^1 \int_{-1}^1 (x+y-1) dx dy \\ &= \int_{-1}^1 (x^2/2 + xy - x) \Big|_{-1}^1 dy = \int_{-1}^1 ((1/2 + y - 1) - (1/2 - y + 1)) dy = \int_{-1}^1 (2y - 2) dy \\ &= (y^2 - 2y) \Big|_{-1}^1 = (1-2) - (1+2) = \underline{-4} \end{aligned}$$

$$\begin{aligned} \text{Hence } \iint_S (x+y+z) dS &= \iint_{S_1} (x+y+z) dS + \iint_{S_2} (x+y+z) dS + \dots + \iint_{S_6} (x+y+z) dS \\ &= 4 + (-4) + 4 + (-4) + 4 + (-4) = 0 \end{aligned}$$

Surface Integral of a vector field over ~~an~~ oriented surface. 15-

Let $\vec{F}$ be a continuous vector field $\vec{F}$ on S be a orientable surface, then the normal surface integral of $\vec{F}$ over S is denoted by $\iint_S \vec{F} \cdot \vec{n} \, dS$ --- (x)

To evaluate (x), it is convenient to express them as double integrals taken over the orthogonal projection of S on one of the coordinate planes.

Evaluation of surface integral
Let R be the orthogonal projection of S on the xy -plane (i.e. S is z -surface)

& let $\vec{n} = \frac{-x\vec{i} - y\vec{j} + z\vec{k}}{\sqrt{x^2 + y^2 + z^2}}$ be unit normal vector directed upward

& projection of dS on the xy plane: $= \sqrt{x^2 + y^2 + z^2} \, dA$

$\Rightarrow$ the integral (x), reduces to the form

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \iint_R \vec{F} \cdot \left(\frac{-x\vec{i} - y\vec{j} + z\vec{k}}{\sqrt{x^2 + y^2 + z^2}} \right) \sqrt{x^2 + y^2 + z^2} \, dA = \iint_R \vec{F} \cdot (-x\vec{i} - y\vec{j} + z\vec{k}) \, dA$$

If $\vec{F} = f\vec{i} + g\vec{j} + h\vec{k}$, then

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \iint_R (f\vec{i} + g\vec{j} + h\vec{k}) \cdot (-x\vec{i} - y\vec{j} + z\vec{k}) \, dA = \iint_R (-fx - gy + hz) \, dA$$

Similarly, if S be a surface with equation $S: y = f(x, z)$ & let R be its projection on the xz -plane, then

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \iint_R \vec{F} \cdot (-x\vec{i} + y\vec{j} - z\vec{k}) \, dA$$

if S be a surface with equation $S: x = f(y, z)$, oriented by normal vector directed in the +ve x -axis & let R be its orthogonal projection on the yz -plane, then

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \iint_R \vec{F} \cdot (x\vec{i} - y\vec{j} - z\vec{k}) \, dA$$

1. Evaluate $\iint_S F \cdot \hat{n} ds$

a) $F(x, y, z) = 2xi + 2yj + zk$; S is the part of the paraboloid $z = 9 - x^2 - y^2$ above the xy -plane; n is directed upward

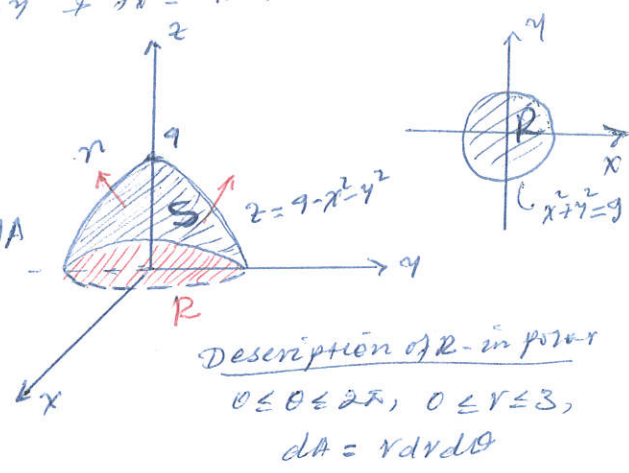
Soln $S: z = 9 - x^2 - y^2$ & the unit normal vector directed upward
 is $\hat{n} = \frac{-2xi - 2yj + k}{\sqrt{4x^2 + 4y^2 + 1}}$, where $f(x, y) = 9 - x^2 - y^2$ & $f_x = -2x$, $f_y = -2y$
 $ds = \sqrt{4x^2 + 4y^2 + 1} dA$

$$\Rightarrow \iint_S F \cdot \hat{n} ds = \iint_R (2xi - 2yj + zk) \cdot \left(\frac{-2xi - 2yj + k}{\sqrt{4x^2 + 4y^2 + 1}} \right) \sqrt{4x^2 + 4y^2 + 1} dA$$

$$= \iint_R (2xi - 2yj + zk) \cdot (-2xi - 2yj + k) dA$$

$$= \iint_R (-4x^2 - 4y^2 + 2z) dA = \iint_R 8 dA$$

$$= \int_0^{2\pi} \int_0^3 8 r dr d\theta = 8 \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^3 d\theta = 8 \int_0^{2\pi} \frac{9}{2} d\theta = 36 \int_0^{2\pi} d\theta = 72\pi$$



b) $F(x, y, z) = -yi + xj + zk$; S is the sphere $x^2 + y^2 + z^2 = 4$; n is directed outward from the sphere.

Soln $S: x^2 + y^2 + z^2 = 4$. A sphere is composed of two hemispheres, each of which is the graph of a function.
 Let S_1 be the upper hemisphere
 S_2 be the lower hemisphere

$$\Rightarrow S_1: z = \sqrt{4 - x^2 - y^2} = f(x, y) \Rightarrow f_x = \frac{-x}{\sqrt{4 - x^2 - y^2}}, f_y = \frac{-y}{\sqrt{4 - x^2 - y^2}}$$

$$\hat{n}_1 = \frac{-fxi - fyj + k}{\sqrt{f_x^2 + f_y^2 + 1}} \quad \& \quad dS = \sqrt{f_x^2 + f_y^2 + 1} dA$$

- directed upward

$$\& S_2: z = -\sqrt{4 - x^2 - y^2} = f(x, y) \Rightarrow f_x = \frac{x}{\sqrt{4 - x^2 - y^2}}, f_y = \frac{y}{\sqrt{4 - x^2 - y^2}}$$

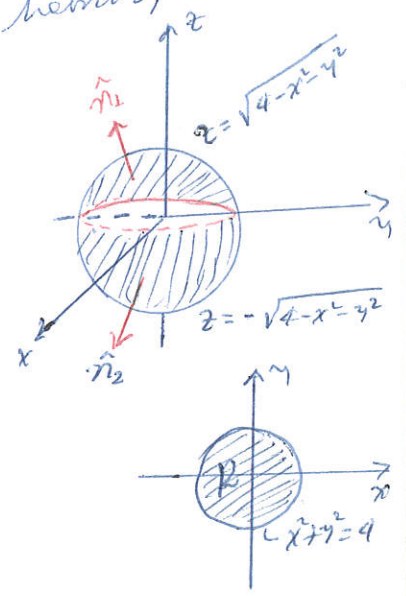
$$\hat{n}_2 = \frac{fxi + fyj - k}{\sqrt{f_x^2 + f_y^2 + 1}} \quad \text{- directed downward}$$

$$\Rightarrow \iint_{S_1} F \cdot \hat{n}_1 ds = \iint_R (-yi + xj + z^2k) \cdot \frac{(-fxi - fyj + k)}{\sqrt{f_x^2 + f_y^2 + 1}} \sqrt{f_x^2 + f_y^2 + 1} dA$$

$$= \iint_R (-yi + xj + z^2k) \cdot (-fxi - fyj + k) dA$$

$$= \iint_R (-yi + xj + z^2k) \cdot \left(\frac{xi}{\sqrt{4 - x^2 - y^2}} + \frac{yj}{\sqrt{4 - x^2 - y^2}} + k \right) dA$$

$$= \iint_R \left(\frac{-yx}{\sqrt{4 - x^2 - y^2}} + \frac{xy}{\sqrt{4 - x^2 - y^2}} + z^2 \right) dA$$



$$\begin{aligned}
 &= \iint_R z^2 dA = \iint_R (4-x^2-y^2)^2 dA = \int_0^{2\pi} \int_0^2 (4-r^2)^2 r dr d\theta = \int_0^{2\pi} \int_0^2 (16-8r^2+r^4) r dr d\theta \\
 &= \int_0^{2\pi} \int_0^2 (16r-8r^3+r^5) dr d\theta = \int_0^{2\pi} \left(8r^2 - 2r^4 + \frac{r^6}{6} \right) \Big|_0^2 d\theta \\
 &= \int_0^{2\pi} \left(32 - 32 + \frac{64}{6} \right) d\theta = \frac{32}{3} \int_0^{2\pi} d\theta = \frac{64\pi}{3} //
 \end{aligned}$$

$$\begin{aligned}
 4 \iint_{S_2} F \cdot \hat{n} ds &= \iint_R (-4i + xj + z^2k) \cdot \frac{(4xi + 4yj - k)}{\sqrt{4x^2+4y^2+1}} dA \\
 &= \iint_R (-4i + xj + z^2k) \cdot (4xi + 4yj - k) dA = \iint_R (-4i + xj + z^2k) \cdot \left(\frac{4xi + 4yj - k}{\sqrt{4-x^2-y^2}} \right) dA \\
 &= \iint_R \left(\frac{-4xy}{\sqrt{4-x^2-y^2}} + \frac{xy}{\sqrt{4-x^2-y^2}} - z^4 \right) dA = - \iint_R z^4 dA = - \iint_R (4-x^2-y^2)^2 dA \\
 &= - \int_0^{2\pi} \int_0^2 (4-r^2)^2 r dr d\theta = - \int_0^{2\pi} \int_0^2 (16-8r^2+r^4) r dr d\theta \\
 &= - \int_0^{2\pi} \int_0^2 (16r-8r^3+r^5) dr d\theta = - \int_0^{2\pi} \left(8r^2 - 2r^4 + \frac{r^6}{6} \right) \Big|_0^2 d\theta \\
 &= - \int_0^{2\pi} \left(32 - 32 + \frac{64}{6} \right) d\theta = - \frac{32}{3} \int_0^{2\pi} d\theta = - \frac{64\pi}{3} //
 \end{aligned}$$

$$\text{Thus } \iint_S F \cdot \hat{n} ds = \iint_{S_1} F \cdot \hat{n} ds + \iint_{S_2} F \cdot \hat{n} ds = \frac{64\pi}{3} - \frac{64\pi}{3} = 0$$

C, $F(x, y, z) = 3x^2i + 6y^2j + z^2k$; S is the closed surface bounded by the cylinder $x^2 + y^2 = 16$, $z=0$, & $z=4$

Soln The surface S contains three parts, S_1 (top), S_2 (bottom),

& S_3 (vertical surface).

$$\iint_S F \cdot \hat{n} ds = \iint_{S_1} F \cdot \hat{n} ds + \iint_{S_2} F \cdot \hat{n} ds + \iint_{S_3} F \cdot \hat{n} ds$$

$$\begin{aligned}
 \text{On } S_1: z=4, \hat{n}=k, \iint_{S_1} F \cdot \hat{n} ds &= \iint_{S_1} z^2 dA = \iint_{S_1} 4 dA \\
 &= 4 \iint_{S_1} dA = 4 (\text{area of a circle of radius 4}) \\
 &= 4 (16\pi) = 64\pi //
 \end{aligned}$$

$$\text{On } S_2: z=0, \hat{n}=-k, \iint_{S_2} F \cdot \hat{n} ds = \iint_{S_2} -z^2 dA = \iint_{S_2} 0 dA = 0$$

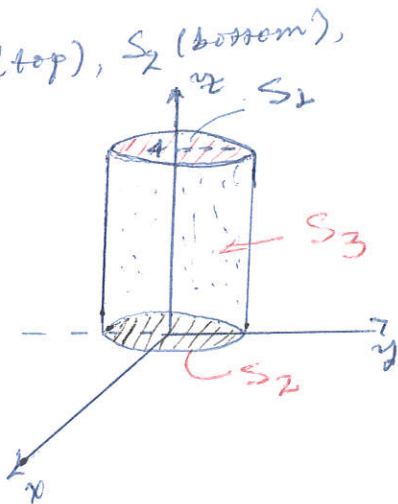
On S_3 : $x^2 + y^2 = 16$, using cylindrical coordinates, we write vector function representation of S_3 as

$$S_3: r(\theta, z) = 4\cos\theta i + 4\sin\theta j + zk, \quad 0 \leq z \leq 4, \quad 0 \leq \theta \leq 2\pi, \quad x = 4\cos\theta, \quad y = 4\sin\theta, \quad z = z$$

$$\begin{aligned}
 \Rightarrow r_\theta &= -4\sin\theta i + 4\cos\theta j + 0k \\
 r_z &= 0i + 0j + k
 \end{aligned}$$

$$\Rightarrow \|r_\theta \times r_z\| = \left\| \begin{vmatrix} i & j & k \\ -4\cos\theta & 4\sin\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} \right\|$$

it consumes our time & energy, & it is avoided even in questions. difficult to consider another method of solving.



$$\begin{aligned}
 & \mathbf{r}_\theta \times \mathbf{r}_z = -4 \cos \theta \mathbf{i} + 4 \sin \theta \mathbf{j} + 0 \mathbf{k} \\
 \Rightarrow \|\mathbf{r}_\theta \times \mathbf{r}_z\| &= \sqrt{(-4 \cos \theta)^2 + (4 \sin \theta)^2 + 0} = 4 \\
 \Rightarrow ds &= \|\mathbf{r}_\theta \times \mathbf{r}_z\| d\theta dz = 4 d\theta dz \\
 \therefore \iint_{S_3} (\mathbf{F} \cdot \mathbf{n}) ds &= \frac{1}{4} \iint_{S_3} (3x^3 + 6y^3) ds
 \end{aligned}$$

$$\begin{aligned}
 S_3: x^2 + y^2 &= 16, \Rightarrow g(x, y, z) = x^2 + y^2 - 16 \\
 \Rightarrow \hat{n} &= \frac{g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}}{\sqrt{g_x^2 + g_y^2 + g_z^2}} = \frac{2x \mathbf{i} + 2y \mathbf{j}}{\sqrt{4x^2 + 4y^2}} \\
 &= \frac{x \mathbf{i} + y \mathbf{j}}{4}, \text{ since } \sqrt{x^2 + y^2} = 4
 \end{aligned}$$

$$R: 0 \leq \theta \leq 2\pi, 0 \leq z \leq 4$$

$$= \frac{1}{4} \iint (3x^3 + 6y^3) 4 d\theta dz$$

$$= \frac{1}{4} \int_0^4 \int_0^{2\pi} [3(4 \cos \theta)^3 + 6(4 \sin \theta)^3] 4 d\theta dz = \int_0^4 \int_0^{2\pi} [192 \cos^3 \theta + 384 \sin^3 \theta] d\theta dz$$

$$= 192 \int_0^4 \int_0^{2\pi} (\cos^3 \theta + 2 \sin^3 \theta) d\theta dz$$

$$= 192 \int_0^4 \int_0^{2\pi} \cos^3 \theta d\theta dz + 384 \int_0^4 \int_0^{2\pi} \sin^3 \theta d\theta dz$$

$$\begin{aligned}
 & \int_0^{2\pi} \cos^3 \theta d\theta = \int_0^{2\pi} \cos^2 \theta \cos \theta d\theta = \int_0^{2\pi} (1 - \sin^2 \theta) \cos \theta d\theta, \text{ let } u = \sin \theta \\
 & \quad du = \cos \theta d\theta \\
 & = \int_0^4 \int_0^{2\pi} (1 - u^2) du dz = \int_0^4 \left(u - \frac{u^3}{3} \right) \Big|_{u=\sin \theta}^{\theta=2\pi} dz = \int_0^4 \left(\sin \theta - \frac{\sin^3 \theta}{3} \right) \Big|_{\theta=0}^{2\pi} dz = 0
 \end{aligned}$$

$$+ \int_0^4 \int_0^{2\pi} \sin^3 \theta d\theta dz = \int_0^4 \int_0^{2\pi} [1 - \cos^2 \theta] \sin \theta d\theta dz = 0$$

$$\text{thus, } \iint_{S_3} \mathbf{F} \cdot \mathbf{n} ds = 0$$

$$\text{Hence, } \iiint_S \mathbf{F} \cdot \mathbf{n} ds = 64\pi + 0 + 0 = \underline{\underline{64\pi}}$$

Divergence Theorem (transformation b/n surface integral & volume integral).

Let D be a simple solid region whose boundary surface S is oriented by the normal $\mathbf{n}$ directed outward from D . If $\mathbf{F}(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$ is a vector field fun having continuous first order partial derivatives in the region D , then

$$\iiint_S \mathbf{F} \cdot \mathbf{n} ds = \iiint_D \text{div } \mathbf{F} dV$$

Eg 1. Evaluate $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, ds$, where $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$; $\nabla \cdot \mathbf{F} = 3$; S is bounded by

$$z = 2 - x^2 - y^2 \text{ and } z = x^2 + y^2 \quad (\hat{\mathbf{n}} \text{ is outward}).$$

Soln $S = S_1 + S_2 \Rightarrow \iint_S = \iint_{S_1} + \iint_{S_2}$

On S_1 : $z = 2 - x^2 - y^2 \Rightarrow \hat{\mathbf{n}} = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$

$$\Rightarrow \iint_{S_1} \mathbf{F} \cdot \hat{\mathbf{n}} \, ds = \iint_R (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \cdot (2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}) \, dA$$

$$= \iint_R (2x^2 + 2y^2 + 1) \, dA$$

Description of R in polar coordinates

$$0 \leq \theta \leq 2\pi, 0 \leq r \leq 1, dA = r \, dr \, d\theta$$

$$\Rightarrow \iint_R (2x^2 + 2y^2 + 1) \, dA = \int_0^{2\pi} \int_0^1 (2r^2 + 1) r \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^1 (2r^3 + r) \, dr \, d\theta = \int_0^{2\pi} \left[\frac{r^4}{2} + \frac{r^2}{2} \right]_0^1 d\theta$$

$$= \int_0^{2\pi} d\theta = \underline{\underline{2\pi}}$$

On S_2 : $z = x^2 + y^2$

$$\Rightarrow \hat{\mathbf{n}} = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}$$

$$\mathbf{F} \cdot \hat{\mathbf{n}} = 2(x^2 + y^2) - 1 \Rightarrow \iint_{S_2} \mathbf{F} \cdot \hat{\mathbf{n}} \, ds = \iint_R (2x^2 + 2y^2 - 1) \, dA$$

$$= \int_0^{2\pi} \int_0^1 (2r^3 - r) \, dr \, d\theta = 0$$

$$\therefore \iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, ds = \iint_{S_1} + \iint_{S_2} = \underline{\underline{2\pi}}$$

or using divergence theorem, $\text{div } \mathbf{F} = 3$.

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, ds = \iiint_V \text{div } \mathbf{F} \, dV = 3 \iiint_V dV = 3 \int_0^{2\pi} \int_0^1 \int_{r^2}^{2-r^2} r \, dz \, dr \, d\theta$$

$$= 3 \int_0^{2\pi} \int_0^1 r [(2-r^2) - r^2] \, dr \, d\theta = 3 \int_0^{2\pi} \int_0^1 (2r - 2r^3) \, dr \, d\theta$$

$$= 3 \int_0^{2\pi} \left[r^2 - \frac{1}{2} r^4 \right]_0^1 d\theta = 3 \cdot \frac{1}{2} \int_0^{2\pi} d\theta = \underline{\underline{3\pi}}$$

Ex 2. Use the divergence theorem to evaluate

$$\iint_S \mathbf{F} \cdot \hat{n} \, ds, \text{ where } \mathbf{F} = x\mathbf{i} - \frac{1}{2}y^2\mathbf{j} + z\mathbf{k} \text{ \& the}$$

surface consists of the surfaces, $z = 4 - 3x^2 - 3y^2$,

$1 \leq z \leq 4$ on the top, $x^2 + y^2 = 1$, $0 \leq z \leq 1$ on the

sides \& $z = 0$ on the bottom.

(S is oriented by a normal vector directed outward)

Soln D is the solid region enclosed

by three surfaces S_1, S_2 \& S_3 .

Using divergence theorem,

$$\iint_S \mathbf{F} \cdot \hat{n} \, ds = \iiint_D \operatorname{div} \mathbf{F} \, dV$$

Description of D

$$0 \leq z \leq 4 - 3r^2$$

$$0 \leq z \leq 1$$

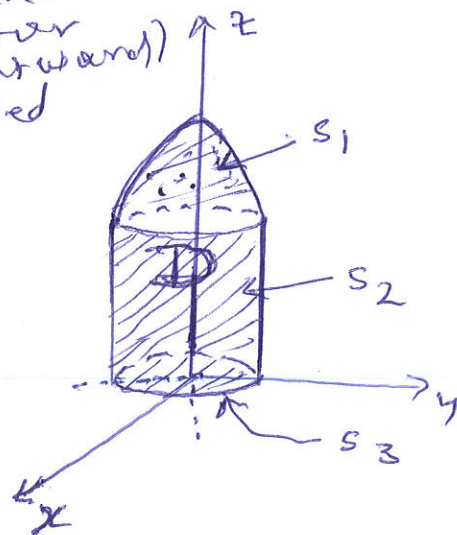
$$0 \leq \theta \leq 2\pi \text{ \& } \operatorname{div} \mathbf{F} = 1 - y + 1 = 1$$

The integral is then, $\iint_S \mathbf{F} \cdot \hat{n} \, ds = \iiint_D \operatorname{div} \mathbf{F} \, dV$

$$= \int_0^{2\pi} \int_0^1 \int_0^{4-3r^2} r \, dz \, dr \, d\theta = \underline{\underline{\frac{5\pi}{2}}}$$

* It would be extremely difficult to evaluate the given surface integral directly. We would have to evaluate 3 - surface integrals corresponding to three pieces of surfaces (S_1, S_2 \& S_3).

* Furthermore, the divergence of $\mathbf{F}$ is much less complicated than $\mathbf{F}$ itself.



Exercise

1. Evaluate $\iint_S F \cdot \hat{n} \, ds$, where

$$F(x, y, z) = 3y^2 \hat{i} + 4x^3 \hat{j} + (2z - x^2) \hat{k},$$

S is bounded by $z = x^2 + y^2$ and $z = 2 - x^2 - y^2$.

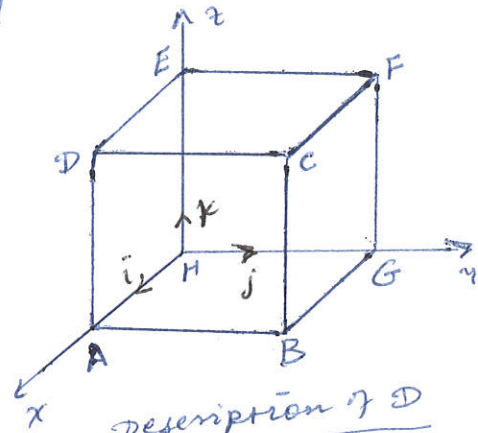
2. Evaluate $\iint_S F \cdot \hat{n} \, ds$, where $F(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$,

S is the surface of the solid bounded by the paraboloid $z = 1 - x^2 - y^2$ and the xy -plane.

Q.1. Verify Divergence theorem for $F(x, y, z) = (x^2 - yz)\mathbf{i} + (y^2 - xz)\mathbf{j} + (z^2 - xy)\mathbf{k}$

taken over the rectangular parallelepiped

$$0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$$



Description of $\mathcal{D}$

$$0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$$

Soln $\text{div} F = 2x + 2y + 2z = 2(x + y + z)$

WTS $\underbrace{\iint_S F \cdot n \, ds}_{\text{LHS}} = \underbrace{\iiint_{\mathcal{D}} \text{div} F \, dV}_{\text{RHS}}$

RHS $\iiint_{\mathcal{D}} \text{div} F \, dV = 2 \int_0^a \int_0^b \int_0^c (x + y + z) \, dz \, dy \, dx$

$$= 2 \int_0^a \int_0^b \left[xz + yz + \frac{z^2}{2} \right]_0^c \, dy \, dx = 2 \int_0^a \int_0^b \left(cx + cy + \frac{c^2}{2} \right) \, dy \, dx$$

$$= 2 \int_0^a \left[cxy + \frac{cy^2}{2} + \frac{c^2}{2} y \right]_0^b \, dx = 2 \int_0^a \left(cbx + \frac{cb^2}{2} + \frac{c^2 b}{2} \right) \, dx$$

$$= 2 \left[\frac{cbx^2}{2} + \frac{cb^2}{2} x + \frac{c^2 b}{2} x \right]_0^a = 2 \left[\frac{cba^2}{2} + \frac{cab^2}{2} + \frac{abc^2}{2} \right]$$

$$= abc(a + b + c) \quad \dots (1)$$

LHS $\iint_S F \cdot n \, ds$ S is composed of six surfaces S_1, S_2, S_3, S_4, S_5 & S_6 where S_1 the face $\square ABCD$, $S_2 \square ABGH$, $S_3 \square EFGH$, S_4 the face $\square BCFG$, S_5 the face $\square CDEF$, S_6 the face $\square ADEH$

$$\Rightarrow \iint_S F \cdot n \, ds = \iint_{S_1} F \cdot n \, ds + \iint_{S_2} F \cdot n \, ds + \dots + \iint_{S_6} F \cdot n \, ds$$

Now, $\iint_{S_1} F \cdot n \, ds$: $n = \mathbf{i}$, $0 \leq y \leq b$, $x = a$, $0 \leq z \leq c$

$$\iint_{S_1} F \cdot n \, ds = \iint_{S_1} F \cdot \mathbf{i} \, ds = \iint_R (x^2 - yz) \, dA = \int_0^b \int_0^c (a^2 - yz) \, dz \, dy = \int_0^b \left(a^2 z - \frac{yz^2}{2} \right) \Big|_0^c \, dy$$

$$= \int_0^b \left(a^2 c - \frac{b^2 z}{2} \right) \, dy = \left(a^2 b z - \frac{b^2}{4} z^2 \right) \Big|_0^c = abc - \frac{b^2 c^2}{4}$$

$\iint_{S_2} F \cdot n \, ds$: $n = -\mathbf{k}$, $z = 0$, $0 \leq x \leq a$, $0 \leq y \leq b$

$$\iint_{S_2} F \cdot n \, ds = \iint_{S_2} F \cdot (-\mathbf{k}) \, ds = \iint_R (xy - z^2) \, dA = \int_0^b \int_0^a (xy) \, dx \, dy = \int_0^b \left[\frac{xy^2}{2} \right]_0^a \, dy = \int_0^b \frac{a^2 y^2}{2} \, dy$$

$$= \frac{a^2}{2} \left[\frac{y^3}{3} \right]_0^b = \frac{a^2 b^3}{6}$$

$\iint_{S_3} F \cdot n \, ds$: $n = \mathbf{i}$, $x = 0$, $0 \leq y \leq b$, $0 \leq z \leq c$

$$\iint_{S_3} F \cdot n \, ds = \iint_{S_3} F \cdot (-\mathbf{i}) \, ds = \iint_R (yz - x^2) \, dA = \int_0^b \int_0^c (yz) \, dz \, dy = \int_0^b \left[\frac{yz^2}{2} \right]_0^c \, dy = \frac{b^2 c^2}{4}$$

On S_4 : $\hat{n} = \hat{j}$, $y = b$, $0 \leq x \leq a$, $0 \leq z \leq c$

$$\begin{aligned} \iint_{S_4} F \cdot n \, ds &= \iint_{S_4} F \cdot (\hat{j}) \, ds = \iint_R (y^2 - zx) \, dA = \int_0^a \int_0^c (b^2 - zx) \, dz \, dx \\ &= \int_0^a \left(b^2 z - \frac{z^2}{2} x \right) \Big|_0^c \, dx = \int_0^a \left(b^2 c - \frac{c^2}{2} x \right) dx = \left(b^2 cx - \frac{c^2}{4} x^2 \right) \Big|_0^a \\ &= b^2 ca - \frac{c^2 a^2}{4} \end{aligned}$$

On S_5 : $\hat{n} = \hat{k}$, $z = c$, $0 \leq x \leq a$, $0 \leq y \leq b$

$$\begin{aligned} \iint_{S_5} F \cdot n \, ds &= \iint_{S_5} F \cdot (\hat{k}) \, ds = \iint_R (z^2 - xy) \, dA = \iint_R (c^2 - xy) \, dA \\ &= \int_0^a \int_0^b (c^2 - xy) \, dy \, dx = \int_0^a \left(c^2 y - x \frac{y^2}{2} \right) \Big|_0^b \, dx \\ &= \int_0^a \left(c^2 b - x \frac{b^2}{2} \right) dx = \left(c^2 bx - \frac{x^2 b^2}{4} \right) \Big|_0^a = c^2 ba - \frac{a^2 b^2}{4} \end{aligned}$$

On S_6 : $\hat{n} = -\hat{j}$, $y = 0$, $0 \leq x \leq a$, $0 \leq z \leq c$

$$\begin{aligned} \iint_{S_6} F \cdot n \, ds &= \iint_{S_6} F \cdot (-\hat{j}) \, ds = \iint_R (zx - y^2) \, dA = \iint_R (zx) \, dA \\ &= \int_0^a \int_0^c zx \, dz \, dx = \int_0^a x \frac{z^2}{2} \Big|_0^c \, dx = \int_0^a \frac{c^2}{2} x \, dx = \left(\frac{c^2 x^2}{4} \right) \Big|_0^a \\ &= \frac{c^2 a^2}{4} // \end{aligned}$$

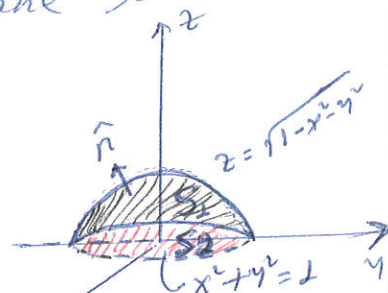
$$\begin{aligned} \text{Thus, } \iiint_S F \cdot n \, ds &= \iint_{S_1} + \iint_{S_2} + \iint_{S_3} + \iint_{S_4} + \iint_{S_5} + \iint_{S_6} \\ &= \cancel{abc} - \cancel{\frac{b^2 c^2}{4}} + \cancel{\frac{a^2 b^2}{4}} + \cancel{\frac{b^2 c^2}{4}} + b^2 ca - \frac{c^2 a^2}{4} + c^2 ba - \cancel{\frac{a^2 b^2}{4}} + \frac{c^2 a^2}{4} \\ &= a^2 bc + b^2 ca + c^2 ba = abc(a + b + c) \quad \dots (2) \end{aligned}$$

From (1) & (2), we see that

$$RHS = LHS$$

& hence Divergence theorem is verified.

Q2 Verify divergence theorem for $F(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ taken over the region bounded by the hemisphere $z = \sqrt{1-x^2-y^2}$ & the disk in the xy -plane bounded by the circle $x^2+y^2=1$.

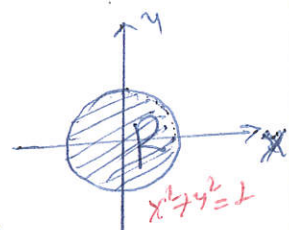


Soln WTS $\iint_S (F \cdot \mathbf{n}) dS = \iiint_D \text{div } F dV$

LHS S is composed of two surfaces $S_1 + S_2$

On S_1 : $z = \sqrt{1-x^2-y^2} = z(x, y) \Rightarrow z_x = \frac{-x}{\sqrt{1-x^2-y^2}}, z_y = \frac{-y}{\sqrt{1-x^2-y^2}}$

$\Rightarrow \iint_{S_1} F \cdot \mathbf{n} dS = \iint_R (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \cdot (-z_x\mathbf{i} - z_y\mathbf{j} + \mathbf{k}) dA$
 $= \iint_R (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \cdot \left(\frac{x\mathbf{i}}{\sqrt{1-x^2-y^2}} + \frac{y\mathbf{j}}{\sqrt{1-x^2-y^2}} + \mathbf{k} \right) dA$



$= \iint_R \left(\frac{x^2}{\sqrt{1-x^2-y^2}} + \frac{y^2}{\sqrt{1-x^2-y^2}} + z \right) dA = \iint_R \left(\frac{x^2+y^2}{\sqrt{1-x^2-y^2}} + \sqrt{1-x^2-y^2} \right) dA$

$= \iint_R \left(\frac{1}{\sqrt{1-x^2-y^2}} \right) dA$

Description of R - in polar C. sy
 $0 \leq \theta \leq 2\pi, 0 \leq r \leq 1, dA = r dr d\theta$

$= \int_0^{2\pi} \int_0^1 \left(\frac{1}{\sqrt{1-r^2}} \right) r dr d\theta$

$= \int_0^{2\pi} \int_0^1 \frac{r dr d\theta}{\sqrt{1-r^2}} + \int \frac{r dr}{\sqrt{1-r^2}}, \text{ let } u = 1-r^2 \Rightarrow -\frac{du}{2} = r dr$

$= \int_0^{2\pi} \int_0^1 \frac{r dr d\theta}{\sqrt{1-r^2}} = -\frac{1}{2} \int \frac{du}{\sqrt{u}} = -\frac{1}{2} \cdot 2\sqrt{u} = -\sqrt{u} = -\sqrt{1-r^2}$

$= \int_0^{2\pi} (-\sqrt{1-r^2}) \Big|_0^1 d\theta$

$= \int_0^{2\pi} (-(\sqrt{1-1} - \sqrt{1-0})) d\theta = \int_0^{2\pi} d\theta = 2\pi$

On S_2 : $\mathbf{n} = -\mathbf{k}, z = 0$ [S_2 is a disk in xy -plane bounded by the circle $x^2+y^2=1$]

$\Rightarrow \iint_{S_2} F \cdot \mathbf{n} dS = \iint_R (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \cdot (-\mathbf{k}) dA = \iint_R (-z) dA = \iint_R 0 dA = 0$

Thus, LHS i.e. $\iint_S F \cdot \mathbf{n} dS = \iint_{S_1} + \iint_{S_2} = 2\pi + 0 = 2\pi \dots (1)$

RHS D is a solid region bounded above by $z = \sqrt{1-x^2-y^2}$ & below by disk $x^2+y^2=1$

Description of D - Rectangular C. sy

$0 \leq z \leq \sqrt{1-x^2-y^2}, -\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}$
 $-1 \leq x \leq 1$

Description in spherical C. sy
 $0 \leq \rho \leq 1, 0 \leq \theta \leq 2\pi$
 $0 \leq \phi \leq \pi/2$

$dV = \rho^2 \sin \phi d\rho d\phi d\theta$

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$$\Rightarrow \iiint_D \text{div } F \, dV = \iiint_D 3 \, dV = 3 \iiint_0^{2\pi} \int_0^{\pi/2} \int_0^1 s^2 \sin \phi \, ds \, d\phi \, d\theta$$

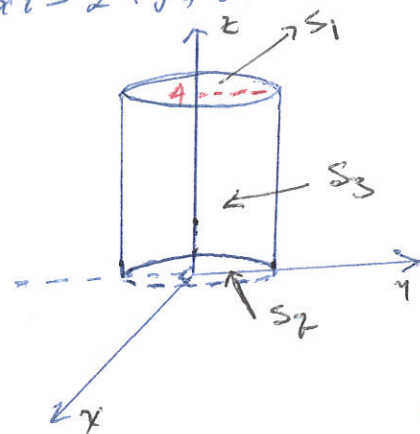
$$= 3 \int_0^{2\pi} \int_0^{\pi/2} \frac{s^3}{3} \Big|_0^1 \sin \phi \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/2} \sin \phi \, d\phi \, d\theta = 3 \int_0^{2\pi} [-\cos \phi]_0^{\pi/2} d\theta$$

$$= \int_0^{2\pi} d\theta = 2\pi \quad \dots (2)$$

Hence the divergence theorem is verified from the equality of (1) & (2).

3. Let D be the region bounded by the cylinder $x^2 + y^2 = 4$, $z=0$, $z=3$. Verify the divergence theorem if $F(x, y, z) = 4x\mathbf{i} - 2y^2\mathbf{j} + z^2\mathbf{k}$.

Soln LHS The surface S contains three parts, S_1 (top), S_2 (bottom) & S_3 (vertical face)



on S_1 : $z=3$, $\hat{n} = \hat{k}$

$$\Rightarrow \iint_{S_1} F \cdot \hat{n} \, dS = \iint_{S_1} z^2 \, dA = \iint_{S_1} 9 \, dA = 9 \iint_{S_1} dA$$

$$= 9 (\text{area of a circle of radius 2})$$

$$= 9 \times 4\pi = \underline{36\pi}$$

on S_2 : $z=0$, $\hat{n} = -\hat{k}$, $\iint_{S_2} F \cdot \hat{n} \, dS = \iint_{S_2} 0 \, dA = 0$

on S_3 : $x^2 + y^2 = 4$ is not simple surface & $\hat{n} = \frac{2x\mathbf{i} + 2y\mathbf{j}}{2\sqrt{x^2 + y^2}} = \frac{x\mathbf{i} + y\mathbf{j}}{2}$
(i.e. $g(x, y, z) = x^2 + y^2 - 4$)

Vector form representation of S_3 using cylindrical coordinates is

$$S_3: \mathbf{r}(\theta, z) = 2\cos\theta\mathbf{i} + 2\sin\theta\mathbf{j} + z\mathbf{k}, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq 3$$

$$\Rightarrow \mathbf{r}_\theta = -2\sin\theta\mathbf{i} + 2\cos\theta\mathbf{j} + 0\mathbf{k} \Rightarrow \mathbf{r}_\theta \times \mathbf{r}_z = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2\sin\theta & 2\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = 2\cos\theta\mathbf{i} + 2\sin\theta\mathbf{j}$$

$$\mathbf{r}_z = 0\mathbf{i} + 0\mathbf{j} + \mathbf{k}$$

$$\Rightarrow \|\mathbf{r}_\theta \times \mathbf{r}_z\| = \sqrt{4} = 2 \quad dA = d\theta \, dz$$

$$\Rightarrow \iint_{S_3} F \cdot \hat{n} \, dS = \iint_{S_3} (4x\mathbf{i} - 2y^2\mathbf{j} + z^2\mathbf{k}) \cdot \left(\frac{x\mathbf{i} + y\mathbf{j}}{2} \right) (2) \, d\theta \, dz$$

$$= \int_0^3 \int_0^{2\pi} [16\cos^2\theta - 16\sin^3\theta] \, d\theta \, dz = 16 \int_0^3 \int_0^{2\pi} \left[\frac{(4 - \cos 2\theta)}{2} - (1 - \cos^2\theta) \sin \theta \right] \, d\theta \, dz$$

$$= 16 \int_0^3 \left[\left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right] + \left[\cos \theta - \frac{\cos^3 \theta}{3} \right] \right]_0^{2\pi} \, dz = 16\pi \int_0^3 dz = 16\pi \times 3 = 48\pi$$

Thus, $\iint_S F \cdot \hat{n} \, dS = \iint_{S_1} + \iint_{S_2} + \iint_{S_3} = 36\pi + 0 + 48\pi = 84\pi \quad \dots (3)$

RHS $\text{div } F = 4 - 4\gamma + 2z$

Description of D in cylindrical

$$0 \leq z \leq 3, 0 \leq \theta \leq 2\pi, 0 \leq r \leq 2$$

$$dV = r dr d\theta dz$$

$$\iiint_D \text{div } F dV = \int_0^3 \int_0^{2\pi} \int_0^2 (4 - 4r + 2z) r dr d\theta dz$$

$$= \int_0^3 \int_0^{2\pi} \int_0^2 (4 - 4r + 2z) r dr d\theta dz$$

$$= \int_0^3 \int_0^{2\pi} (4 - 8\sin\theta + 2z) \frac{r^2}{2} \Big|_0^2 d\theta dz = 2 \int_0^3 \int_0^{2\pi} (4 - 8\sin\theta + 2z) d\theta dz$$

$$= 2 \int_0^3 \int_0^{2\pi} (4 - 8\sin\theta + 2z) d\theta dz = 2 \int_0^3 (4\theta + 8\cos\theta + 2z\theta) \Big|_0^{2\pi} dz$$

$$= 2 \int_0^3 (8\pi + 2\pi z) dz = 2 \int_0^3 (8\pi + \frac{2\pi z^2}{2}) \Big|_0^3 dz$$

$$= 2(24\pi + 18\pi) = 2(42\pi) = \underline{\underline{84\pi}} \dots (2)$$

$\Rightarrow$ LHS = RHS & hence divergence theorem is verified.

4. Use divergence theorem to compute $\iint_S F \cdot n ds$, where n is normal to S that is directed outward.

a. $F(x, y, z) = -2x\mathbf{i} + 4y\mathbf{j} - 7z\mathbf{k}$; S is the boundary of the solid region from above & below by sphere $x^2 + y^2 + z^2 = 4$ & on the sides by the cylinder $x^2 + y^2 = 1$.

b. $F(x, y, z) = yz\mathbf{i} + xy\mathbf{j} + xz\mathbf{k}$; S is the boundary of the solid region inside the cylinder $x^2 + y^2 = 1$ & b/n the planes $y = -1$ & $y = 1$.

1/2 a. $F(x, y, z) = -2x\mathbf{i} + 4y\mathbf{j} - 7z\mathbf{k} \Rightarrow \text{div } F = -2 + 4 - 7 = -5$

$$\iint_S F \cdot n ds = \iiint_D \text{div } F dV$$

Description of D in cyl. c. sy

$$-\sqrt{4-x^2-y^2} \leq z \leq \sqrt{4-x^2-y^2}$$

$$0 \leq r \leq 1, 0 \leq \theta \leq 2\pi$$

$$\text{i.e. } x^2 + y^2 = r^2$$

$$\Rightarrow -\sqrt{4-r^2} \leq z \leq \sqrt{4-r^2}$$

$$= \int_0^{2\pi} \int_0^1 \int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} (-5) r dz dr d\theta$$

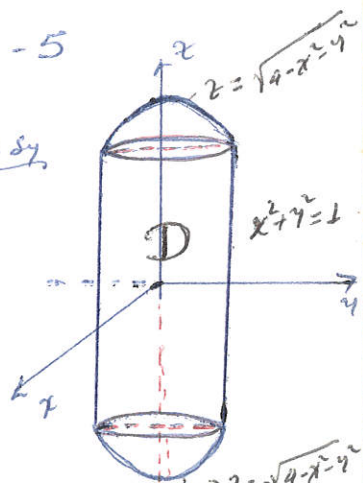
$$= -5 \int_0^{2\pi} \int_0^1 2r \sqrt{4-r^2} dr d\theta = -5 \int_0^{2\pi} \int_0^1 r 2\sqrt{4-r^2} dr d\theta \quad dV = r dz dr d\theta$$

$$= -5 \int_0^{2\pi} \int_0^1 2r \sqrt{4-r^2} dr d\theta = -10 \int_0^{2\pi} \int_0^1 r \sqrt{4-r^2} dr d\theta \quad \text{let } u = 4-r^2$$

$$= -10 \int_0^{2\pi} \left[-\frac{1}{4} \frac{1}{\sqrt{4-r^2}} \right]_0^1 d\theta = \frac{5}{2} \int_0^{2\pi} \left(\frac{1}{\sqrt{3}} - \frac{1}{2} \right) d\theta \quad -\frac{1}{2} \int \sqrt{u} du = -\frac{1}{2} \cdot \frac{1}{2} \sqrt{4-r^2}$$

$$= \frac{5}{2} \left(\frac{1}{\sqrt{3}} - \frac{1}{2} \right) 2\pi = 5\pi \left(\frac{2-\sqrt{3}}{2\sqrt{3}} \right)$$

b. exercise



5. Verify divergence theorem for $F(x, y, z) = xi + yj + zk$ taken over the region bounded ~~by the sides~~ ^{on the} by the cylinder $x^2 + y^2 = 4$, above by the plane $z = y + 4$, & below by $z = y - 4$.

Soln S is composed of the surfaces S_1, S_2 & S_3

$$S_1: z = y + 4, \hat{n} = \frac{-j + k}{\sqrt{2}} = \frac{1}{\sqrt{2}}(-j + k)$$

$$dS = \sqrt{2} dA$$

$$\iint_{S_1} F \cdot n ds = \iint_R (xi + yj + zk) \cdot \frac{1}{\sqrt{2}}(-j + k) \sqrt{2} dA$$

$$= \iint_R (-y + z) dA = \iint_R (-y + y + 4) dA$$

$$= 4 \iint_R dA = 4 (\text{area of the circle of radius 2}), R \text{ is projection of } S_1 \text{ into } xy\text{-plane}$$

$$= 16\pi$$

$$\text{On } S_2: z = y - 4 \Rightarrow \hat{n} = \frac{j - k}{\sqrt{2}} = \frac{1}{\sqrt{2}}(j - k) \quad dS = \sqrt{2} dA$$

$$\Rightarrow \iint_{S_2} (xi + yj + zk) \cdot \frac{1}{\sqrt{2}}(j - k) \sqrt{2} dA = \iint_R (xi + yj + zk) \cdot (j - k) dA$$

$$= \iint_R (y - z) dA = \iint_R (y - y + 4) dA = 4 \iint_R dA = 4 \cdot 4\pi = 16\pi, R \text{ is proj. of } S_2 \text{ into } xy\text{-plane}$$

On S_3 : S_3 is represented parametrically by a vector fun

$$r(\theta, z) = 2\cos\theta i + 2\sin\theta j + zk, \quad 0 \leq z \leq 2\pi, \quad y - 4 \leq z \leq y + 4$$

$$+ \|r_\theta \times r_z\| = 2 \quad \& \quad ds = 2 d\theta dz = 2 dz d\theta$$

$$\& \hat{n} = \frac{2xi + 2yj}{2\sqrt{x^2 + y^2}} = \frac{xi + yj}{2} = \frac{1}{2}(xi + yj)$$

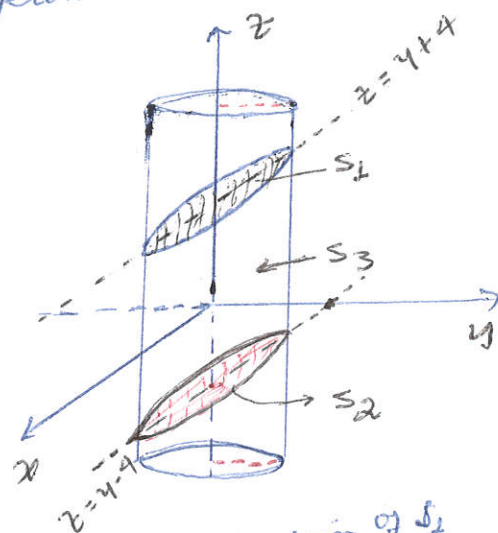
$$\Rightarrow \iint_{S_3} F \cdot n ds = \iint_R (xi + yj + zk) \cdot \frac{1}{2}(xi + yj) 2 dz d\theta = \iint_R (x^2 + y^2) dz d\theta$$

$$= \int_0^{2\pi} \int_{2\sin\theta - 4}^{2\sin\theta + 4} 4 dz d\theta = 4 \int_0^{2\pi} z \Big|_{2\sin\theta - 4}^{2\sin\theta + 4} d\theta = 4 \int_0^{2\pi} (2\sin\theta + 4 - 2\sin\theta + 4) d\theta$$

$$= 4 \int_0^{2\pi} 8 d\theta = 32\theta \Big|_0^{2\pi} = 64\pi$$

$$\text{Thus, } \iint_S F \cdot n ds = \iint_{S_1} F \cdot n ds + \iint_{S_2} F \cdot n ds + \iint_{S_3} F \cdot n ds$$

$$= 32\pi + 64\pi = 96\pi$$



$$\operatorname{div} F = 1+1+1=3$$

$$\Rightarrow \iiint_D \operatorname{div} F dV = \iiint_D 3 dV = 3 \iiint_D dV$$

$$= 3 \iiint_D dV = 3 \int_0^{2\pi} \int_0^2 \int_{r \sin \theta - 4}^{r \sin \theta + 4} r dz dr d\theta$$

$$= 3 \int_0^{2\pi} \int_0^2 r(z) \Big|_{r \sin \theta - 4}^{r \sin \theta + 4} dr d\theta = 3 \int_0^{2\pi} \int_0^2 r(r \sin \theta + 4 - r \sin \theta + 4) dr d\theta$$

$$= 3 \int_0^{2\pi} \int_0^2 8r dr d\theta = 12 \int_0^{2\pi} r^2 \Big|_0^2 d\theta = 12 \times 4 \int_0^{2\pi} d\theta = 48 \cdot \theta \Big|_0^{2\pi} = 96\pi$$

which agrees with the value of surface integral obtained above

Description of D - in cylindrical co. sys

$$0 \leq \theta \leq 2\pi, 0 \leq r \leq 2$$

$$r \sin \theta - 4 \leq z \leq r \sin \theta + 4$$

$$dV = r dz dr d\theta$$

Stokes' Theorem

Let S be an oriented smooth surface that is bounded by a simple closed, smooth boundary curve C with positive orientation and further let F be any continuously differentiable vector field then

$$\int_C F \cdot \hat{n} \, ds = \iint_S \text{curl } F \cdot \hat{n} \, ds, \text{ where the surface}$$

S can be any surface so long as its boundary curve is given by C .

Stokes's Theorem (transformation b/w line & surface integral)

If S be an open surface bounded by a closed curve C oriented in the counter clockwise by a position $\vec{r}(u, v)$ & let $\vec{F}(x, y, z) = f(x, y, z)\vec{i} + g(x, y, z)\vec{j} + h(x, y, z)\vec{k}$ be any continuously differentiable vector function, then

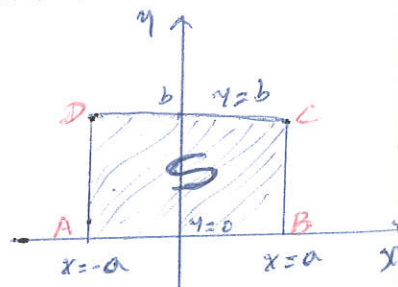
$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds, \text{ where } \hat{n} \text{ is the unit normal of } S \text{ and } \vec{r}(u, v) \text{ is the parametric representation of a curve } C \text{ that is oriented in the counter clockwise (the direction).}$$

$$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

Q1. Verify Stokes's theorem for $\vec{F}(x, y, z) = (x^2 + y^2)\vec{i} - 2xy\vec{j}$ taken around the rectangle bounded by the line $x = \pm a, y = 0, y = b$

Soln WTS. $\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$

LHS Let $ABCD = C$ be the given rectangle, as shown here by here, C is composed of four line segments $\overline{AB}, \overline{BC}, \overline{CD}, \text{ \& } \overline{DA}$



$$\Rightarrow \int_C \vec{F} \cdot d\vec{r} = \int_{AB} \vec{F} \cdot d\vec{r} + \int_{BC} \vec{F} \cdot d\vec{r} + \int_{CD} \vec{F} \cdot d\vec{r} + \int_{DA} \vec{F} \cdot d\vec{r}$$

on AB: $y = 0, -a \leq x \leq a, dy = 0$

$$\Rightarrow \int_{AB} \vec{F} \cdot d\vec{r} = \int_{AB} ((x^2 + y^2)\vec{i} - 2xy\vec{j}) \cdot (dx\vec{i} + dy\vec{j}) = \int_{AB} (x^2 + y^2) dx = \int_{-a}^a x^2 dx = \left. \frac{x^3}{3} \right|_{-a}^a = \frac{2a^3}{3}$$

on BC: $x = a, dx = 0, 0 \leq y \leq b$

$$\int_{BC} \vec{F} \cdot d\vec{r} = \int_{BC} ((x^2 + y^2)\vec{i} - 2xy\vec{j}) \cdot (dx\vec{i} + dy\vec{j}) = \int_0^b -2ay dy = -2a \int_0^b y dy = -ab^2$$

on CD: $y = b, dy = 0, a \leq x \leq -a$

$$\int_{CD} \vec{F} \cdot d\vec{r} = \int_{CD} ((x^2 + y^2)dx - 2xydy) = \int_a^{-a} (x^2 + b^2) dx = \left. \frac{x^3}{3} + xb^2 \right|_a^{-a} = -\frac{a^3}{3} - ab^2 - \left(\frac{a^3}{3} + ab^2 \right) = -\frac{2a^3}{3} - 2ab^2$$

on DA : $x = -a$, $dx = 0$, $b \leq y \leq 0$

$$\int_C F \cdot dr = \int_{DA} (x^2 + y^2) dx - 2xy dy = \int_b^0 (a^2 + y^2) 0 + 2ay dy = 2 \int_b^0 ay dy$$

$$= 2a \int_b^0 y dy = 2a \left[\frac{y^2}{2} \right]_b^0 = -ab^2$$

Thus, $\int_C F \cdot dr = \int_{ABCD} F \cdot dr = \frac{2a^3}{3} - ab^2 - \frac{2a^3}{3} - 2ab^2 - ab^2 = -4ab^2$

RHS: since $\text{curl } F = -4yz \mathbf{k} =$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y^2 & -2xy & 0 \end{vmatrix}$$

and the unit normal vector of the rectangle ABCD is

$\hat{n} = \mathbf{k}$

$$\therefore \iint_S \text{curl } F \cdot \hat{n} ds = \int_0^b \int_{-a}^a (-4yz) \cdot \mathbf{k} \cdot \mathbf{k} dx dy = -4 \int_0^b \int_{-a}^a y dx dy = -4 \int_0^b y(x) \Big|_{-a}^a dy$$

$$= (-4)(2a) \int_0^b y dy = -8a \left[\frac{y^2}{2} \right]_0^b = -\frac{8ab^2}{2} = -4ab^2$$

Hence Stokes's theorem is verified from equality of LHS & RHS.

Q2. Let S be the portion of the paraboloid $z = 4 - x^2 - y^2$ for which $z \geq 0$, & let C be the circle $x^2 + y^2 = 4$ that forms the boundary of S in the xy -plane. Verify Stokes's theorem for the vector field $F(x, y, z) = 2z\mathbf{i} + 3x\mathbf{j} + 5y\mathbf{k}$ if S is oriented upward.

Soln C can be represented parametrically by

LHS $x = 2\cos t$, $y = 2\sin t$, $z = 0$, $(0 \leq t \leq 2\pi)$

i.e. $C: r(t) = 2\cos t \mathbf{i} + 2\sin t \mathbf{j}$, $0 \leq t \leq 2\pi$

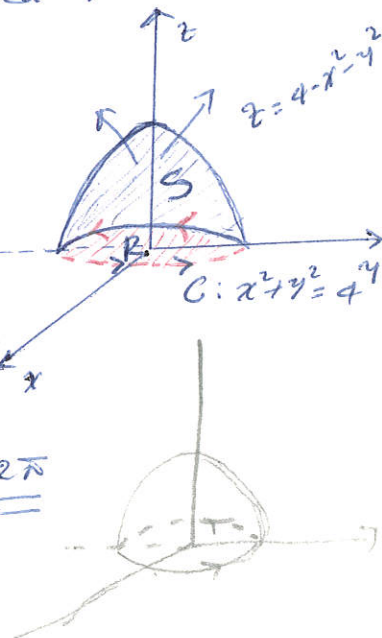
Therefore, $\int_C F \cdot dr = \int_C (2z\mathbf{i} + 3x\mathbf{j} + 5y\mathbf{k}) \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k})$

$$= \int_C 2z dx + 3x dy = \int_0^{2\pi} [2(0)(-2\sin t) + 3(2\cos t)(2\cos t) dt]$$

$$= 12 \int_0^{2\pi} \cos^2 t dt = 12 \left[\frac{t}{2} + \frac{\sin 2t}{4} \right]_0^{2\pi} = 12 \cdot \frac{2\pi}{2} = 12\pi$$

RHS: To evaluate the right side, we note that

$$\text{curl } F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2z & 3x & 5y \end{vmatrix} = 5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$$



since S is oriented by a normal vector directed upward & is represented in the form $z = 4 - x^2 - y^2 = g(x, y)$, then $\hat{n}$ is given by

$$\hat{n} = \frac{-g_x \mathbf{i} - g_y \mathbf{j} + \mathbf{k}}{\sqrt{g_x^2 + g_y^2 + 1}} = \frac{2x \mathbf{i} + 2y \mathbf{j} + \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}} \quad \& \quad dS = \sqrt{4x^2 + 4y^2 + 1} \, dA$$

$$\Rightarrow \iint_S \text{Curl } \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R (5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) \cdot \frac{(2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k})}{\sqrt{4x^2 + 4y^2 + 1}} \sqrt{4x^2 + 4y^2 + 1} \, dA$$

$$= \iint_R (10x + 4y + 3) \, dA$$

$$= \int_0^{2\pi} \int_0^2 (10r \cos \theta + 4r \sin \theta + 3) r \, dr \, d\theta \quad \text{Description of } D \text{ in polar coordinate } r, \theta.$$

$$= \int_0^{2\pi} \int_0^2 (10r^2 \cos \theta + 4r^2 \sin \theta + 3r) \, dr \, d\theta \quad \begin{matrix} 0 \leq \theta \leq 2\pi, 0 \leq r \leq 2 \\ dA = r \, dr \, d\theta \end{matrix}$$

$$= \int_0^{2\pi} \left(10r^3 \cos \theta + 4r^3 \sin \theta + \frac{3}{2} r^2 \right) \Big|_0^2 \, d\theta = \int_0^{2\pi} \left(\frac{80}{3} \cos \theta + \frac{32}{3} \sin \theta + 6 \right) \, d\theta$$

$$= \frac{80}{3} (\sin \theta) \Big|_0^{2\pi} - \frac{32}{3} \cos \theta \Big|_0^{2\pi} + 6\theta \Big|_0^{2\pi} = \frac{80}{3} \cdot 0 - \frac{32}{3} (\cos 2\pi - \cos 0) + 12\pi$$

$$= 12\pi$$

Thus, LHS = RHS & hence Stokes's theorem is verified.

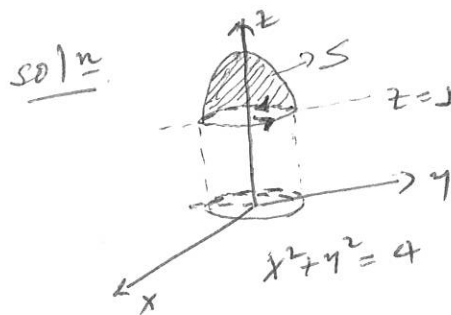
Exercise 1. Verify Stokes's theorem for the vector field

$\mathbf{F}(x, y, z) = (2x - y)\mathbf{i} - yz^2\mathbf{j} - y^2z\mathbf{k}$ over the upper half surface of $x^2 + y^2 + z^2 = 1$, bounded by its projection on the xy -plane.

2. Verify Stokes's theorem for $\mathbf{F}(x, y, z) = (y - z + 2)\mathbf{i} + (yz + 4)\mathbf{j} - xz\mathbf{k}$ where S is the surface of the cube $x=0, y=0, z=0, x=2, y=2, z=2$ above the xy -plane.

3. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y, z) = 4z\mathbf{i} + 5xy\mathbf{j} + 6y\mathbf{k}$, C is the boundary of the portion of paraboloid $z = 1 - x^2 - y^2$ above the plane $z=0$ (oriented counter clockwise).

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 Ex 1. Evaluate $\int_C F \cdot dr$, where $F(x, y, z) = 2z\mathbf{i} + 3x\mathbf{j} + 5y\mathbf{k}$, C is the boundary of the portion of paraboloid $z = 5 - x^2 - y^2$ above $z = 1$ (oriented counter clockwise)



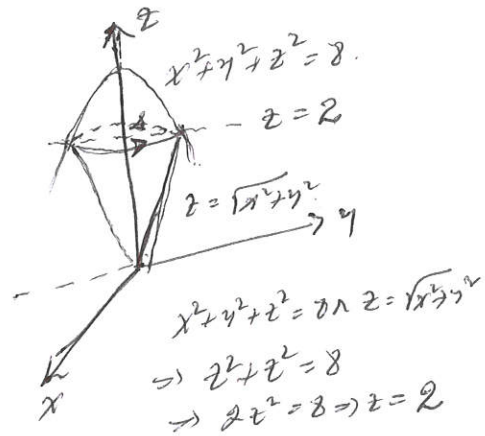
C - can be represented parametrically by
 $x = 2\cos t, y = 2\sin t, z = 1, (0 \leq t \leq 2\pi)$
 $(C: r(t)) = 2\cos t\mathbf{i} + 2\sin t\mathbf{j} + \mathbf{k}, 0 \leq t \leq 2\pi$

$$\begin{aligned} \therefore \int_C F \cdot dr &= \int_0^{2\pi} (4\cos t\mathbf{i} + 6\cos t\mathbf{j} + 10\sin t\mathbf{k}) \cdot (-2\sin t\mathbf{i} + 2\cos t\mathbf{j} + 0\mathbf{k}) dt \\ &= \int_0^{2\pi} (-48\sin t + 12\cos^2 t) dt \\ &= 4\cos t \Big|_0^{2\pi} + 12 \left(\frac{t}{2} + \frac{\sin 2t}{4} \right) \Big|_0^{2\pi} = 0 + 12(\pi) \\ &= 12\pi \end{aligned}$$

Using surface integral, $\text{curl } F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2z & 3x & 5y \end{vmatrix} = 5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$

$$\begin{aligned} \int_C F \cdot dr &= \iint_S \text{curl } F \cdot \mathbf{n} \, dS = \iint_S (5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) \cdot (2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}) \, dA \\ &= \iint_S (10x + 4y + 3) \, dA = \int_0^{2\pi} \int_0^2 (10r\cos\theta + 2r\sin\theta + 3)r \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^2 (10r^2\cos\theta + 2r^2\sin\theta + 3r) \, dr \, d\theta \\ &= \int_0^{2\pi} \left(\frac{10r^3}{3}\cos\theta + \frac{2r^3}{3}\sin\theta + \frac{3r^2}{2} \right) \Big|_0^2 d\theta \\ &= \int_0^{2\pi} \left(\frac{80}{3}\cos\theta + \frac{16}{3}\sin\theta + 6 \right) d\theta \\ &= \frac{80}{3}\sin\theta - \frac{16}{3}\cos\theta + 6\theta \Big|_0^{2\pi} = 0 + 0 + 12\pi \\ &= 12\pi \end{aligned}$$

2. $\int_C F \cdot d\mathbf{r}$, where C is the intersection of the sphere $x^2 + y^2 + z^2 = 8$ & the cone $z = \sqrt{x^2 + y^2}$ with counter clockwise orientation &
 $F(x, y, z) = 2zy\mathbf{i} - 3y\mathbf{j} + yx\mathbf{k}$.



Soln $C: \mathbf{r}(t) = 2\cos t\mathbf{i} + 2\sin t\mathbf{j} + 2\mathbf{k}, 0 \leq t \leq 2\pi$.

$$\int_C F \cdot d\mathbf{r} = \int_0^{2\pi} (2\cos t\mathbf{i} - 6\sin t\mathbf{j} + 4\cos t\sin t\mathbf{k}) \cdot (-2\sin t\mathbf{i} + 2\cos t\mathbf{j} + 0\mathbf{k}) dt$$

$$= \int_0^{2\pi} (-16\sin^2 t - 12\sin t \cos t) dt$$

$$= \left[16 \left(\frac{t}{2} - \frac{\sin 2t}{4} \right) - 12 \frac{\sin^2 t}{2} \right]_0^{2\pi} = -16 \left(\frac{2\pi}{2} \right) = -16\pi$$

Using surface integral, $\text{curl } F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2zy & -3y & yx \end{vmatrix} = \mathbf{i}(x) - \mathbf{j}(y - 2y) + \mathbf{k}(-2z)$

$$= x\mathbf{i} + y\mathbf{j} - 2z\mathbf{k}, \quad \text{since } z = \sqrt{x^2 + y^2}, \quad q_x = \frac{x}{\sqrt{x^2 + y^2}}, \quad q_y = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\Rightarrow \hat{n} = -\frac{x}{\sqrt{x^2 + y^2}}\mathbf{i} - \frac{y}{\sqrt{x^2 + y^2}}\mathbf{j} + \mathbf{k}$$

$$\Rightarrow \iint_S \text{curl } F \cdot \hat{n} \, dA$$

$$= \iint_R (x\mathbf{i} + y\mathbf{j} - 2z\mathbf{k}) \cdot \left(-\frac{x}{\sqrt{x^2 + y^2}}\mathbf{i} - \frac{y}{\sqrt{x^2 + y^2}}\mathbf{j} + \mathbf{k} \right) dA$$

$$= \iint_R \left(-\frac{x^2}{\sqrt{x^2 + y^2}} - \frac{y^2}{\sqrt{x^2 + y^2}} - 2z \right) dA = -2 \iint_R z \, dA = -2 \int_0^{2\pi} \int_0^2 r \, r \, dr \, d\theta$$

$$= -2 \int_0^{2\pi} \left[\frac{r^3}{3} \right]_0^2 d\theta = -2 \int_0^{2\pi} \frac{8}{3} d\theta$$

$$= \iint_R \left(-\frac{x^2 + y^2}{\sqrt{x^2 + y^2}} - 2\sqrt{x^2 + y^2} \right) dA = \iint_R \left(-\frac{r^2}{r} - 2r \right) r \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 -3r^2 \, dr \, d\theta = -\int_0^{2\pi} \left[-r^3 \right]_0^2 d\theta = -8 \int_0^{2\pi} d\theta = (-8)(2\pi)$$

$$= \underline{\underline{-16\pi}}$$

$$\text{or } S: z = \sqrt{8-x^2-y^2} = g(x,y), \quad g_x = \frac{-x}{\sqrt{8-x^2-y^2}}, \quad g_y = \frac{-y}{\sqrt{8-x^2-y^2}}$$

$$\vec{n} = \frac{x}{\sqrt{8-x^2-y^2}} \vec{i} + \frac{y}{\sqrt{8-x^2-y^2}} \vec{j} + \vec{k}$$

$$\Rightarrow \iint_S \text{curl } F \cdot \vec{n} \, dA = \iint_R (xi + yj - 2zk) \cdot \left(\frac{x}{\sqrt{8-x^2-y^2}} \vec{i} + \frac{y}{\sqrt{8-x^2-y^2}} \vec{j} + \vec{k} \right) dA$$

$$= \iint_R \left(\frac{x^2+y^2}{\sqrt{8-x^2-y^2}} - 2z \right) dA = \iint_R \left(\frac{r^2}{\sqrt{8-r^2}} - 2\sqrt{8-r^2} \right) r \, dr \, d\theta$$

$$= \iint_R \left(\frac{r^2 - 2(8-r^2)}{\sqrt{8-r^2}} \right) r \, dr \, d\theta = \iint_R \frac{(3r^2 - 16)r}{\sqrt{8-r^2}} \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 \frac{(3r^3 - 16r)}{\sqrt{8-r^2}} \, dr \, d\theta = \int_0^{2\pi} \left[\int_0^2 \frac{(3r^3 - 16r)}{\sqrt{8-r^2}} \, dr \right] d\theta$$

$$\begin{aligned} u &= 8-r^2 \\ du &= -2r \, dr \\ \frac{du}{-2} &= r \, dr \\ r^2 &= 8-u \end{aligned}$$

$$= -\frac{1}{2} \int_0^{2\pi} \int_0^2 \frac{(3(8-u) - 16)}{\sqrt{u}} \, du \, d\theta = -\frac{1}{2} \int_0^{2\pi} \int_0^2 \frac{(24 - 3u - 16)}{\sqrt{u}} \, du \, d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} \int_0^2 \frac{8-3u}{\sqrt{u}} \, du \, d\theta = -\frac{1}{2} \int_0^{2\pi} \int_0^2 (8u^{-1/2} - 3u^{1/2}) \, du \, d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} \int_0^2 (16u^{1/2} - 2u^{3/2}) \, du \, d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} \left(16(8-r^2)^{1/2} - 2(8-r^2)^{3/2} \right) \Big|_0^2 d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} \left((32-16) - 16(\sqrt{8}) + 2\sqrt{64 \cdot 8} \right) d\theta$$

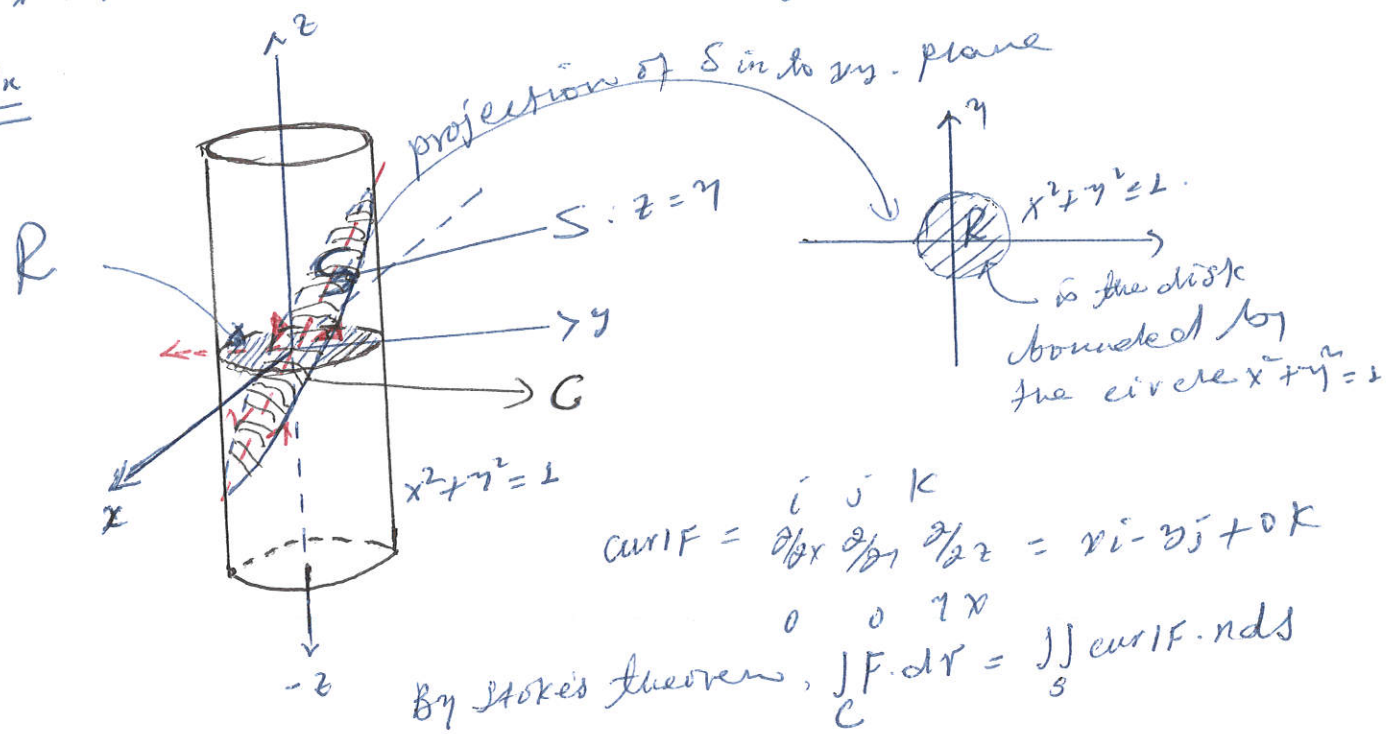
$$= -\frac{1}{2} \int_0^{2\pi} (16 - 16\sqrt{8} + 2\sqrt{64 \cdot 8}) \, d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} (16 - 16\sqrt{8} + 16\sqrt{8}) \, d\theta = -\frac{1}{2} \int_0^{2\pi} 16 \, d\theta$$

$$= -8 \int_0^{2\pi} d\theta = -8\theta \Big|_0^{2\pi} = \underline{\underline{-16\pi}}$$

Q. Let $F(x, y, z) = yxk$, & C is the intersection of the circular $x^2 + y^2 = 1$ & the plane $z = y$ (C is oriented in counter clockwise direction).

Soln



$$\text{for } S: z = y = g(x, y) \Rightarrow ds = \sqrt{1+1} \, dA = \sqrt{2} \, dA$$

$$n = 0i - j + k / \sqrt{2}$$

$$\Rightarrow \iint_S \text{curl } F \cdot n \, ds = \iint_R (xi - yj + 0k) \cdot \frac{(0i - j + k)}{\sqrt{2}} \sqrt{2} \, dA$$

Description of R in polar. Co.

$$0 \leq \theta \leq 2\pi, 0 \leq r \leq 1, dA = r \, dr \, d\theta$$

$$= \iint_R y \, dA = \int_0^{2\pi} \int_0^1 r^2 \sin \theta \, dr \, d\theta$$

$$= \int_0^{2\pi} \frac{r^3}{3} \Big|_0^1 \sin \theta \, d\theta = \frac{1}{3} \int_0^{2\pi} \sin \theta \, d\theta = 0$$

$\Rightarrow$ the right side integral is zero i.e. $\iint_S \text{curl } F \cdot n \, ds = 0$.

$\Rightarrow$ the right side integral is zero i.e. $\iint_S \text{curl } F \cdot n \, ds = 0$.

Next, we need to check that the left handside integral

$\oint_C F \cdot dr$ is also vanishes

parametric Repres. of $S: x^2 + y^2 = 1, z = y$

$$S: r(\theta) = \cos \theta i + \sin \theta j + z k, \text{ but } z = y = \sin \theta \Rightarrow dz = \cos \theta d\theta$$

$$= \cos \theta i + \sin \theta j + \sin \theta k, 0 \leq \theta \leq 2\pi$$

$$\Rightarrow \oint_C F \cdot dr = \int_0^{2\pi} xyk(dx i + dy j + dz k) = \int_0^{2\pi} xy \, dz = \int_0^{2\pi} \cos \theta \sin \theta (+\cos \theta) d\theta$$

$$= \int_0^{2\pi} \cos^2 \theta \sin \theta \, d\theta = -\frac{\cos^3 \theta}{3} \Big|_0^{2\pi} = 0$$

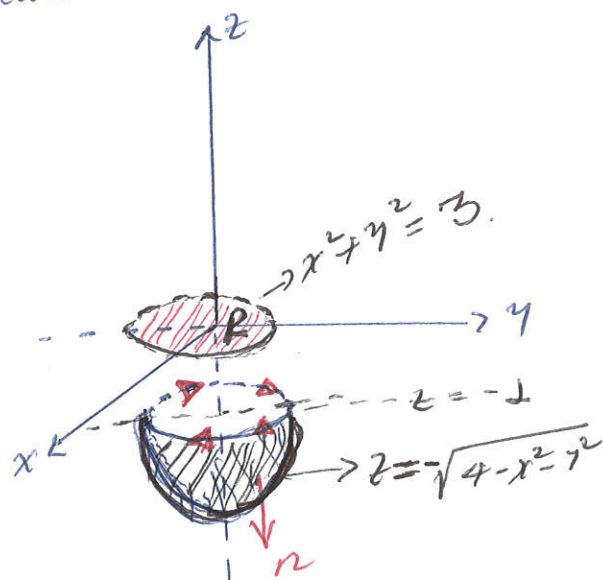
Q. Evaluate $\int_C F \cdot dr$, where $F(x, y, z) = z\mathbf{i} + x\mathbf{j} + y\mathbf{k}$,
 C is the ~~intersection~~ ^{boundary} of the portion of sphere $x^2 + y^2 + z^2 = 4$
 below the plane $z = -1$, oriented in clockwise.

Soln. $\text{curl } F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

$S: z = \sqrt{4 - x^2 - y^2} = g(x, y)$

$\mathbf{n} = \frac{g_x \mathbf{i} + g_y \mathbf{j} - \mathbf{k}}{\sqrt{g_x^2 + g_y^2 + 1}}, \quad g_x = \frac{-x}{\sqrt{4 - x^2 - y^2}}$

$g_y = \frac{-y}{\sqrt{4 - x^2 - y^2}}$



$dS = \sqrt{g_x^2 + g_y^2 + 1} dA$

$\int_C F \cdot dr = \iint_S \text{curl } F \cdot \mathbf{n} dS = \iint_R (\mathbf{i} + \mathbf{j} + \mathbf{k}) \cdot (g_x \mathbf{i} + g_y \mathbf{j} - \mathbf{k}) dA$

$= \iint_R (\mathbf{i} + \mathbf{j} + \mathbf{k}) \cdot \left(\frac{-x\mathbf{i}}{\sqrt{4 - x^2 - y^2}} + \frac{-y\mathbf{j}}{\sqrt{4 - x^2 - y^2}} - \mathbf{k} \right) dA$

$= \iint_R \left(\frac{-x}{\sqrt{4 - x^2 - y^2}} + \frac{-y}{\sqrt{4 - x^2 - y^2}} - 1 \right) dA = \iint_R \left(\frac{-x - y}{\sqrt{4 - x^2 - y^2}} - 1 \right) dA$

$= \iint_R \frac{-x - y}{\sqrt{4 - x^2 - y^2}} dA - \iint_R dA = \int_0^{2\pi} \int_0^{\sqrt{3}} \frac{-r \cos \theta - r \sin \theta}{\sqrt{4 - r^2}} r dr d\theta - \iint_R dA$

$= -3\pi$

OR ~~Case~~ $x^2 + y^2 + z^2 = 4, z = -1 \Rightarrow x^2 + y^2 = 3$

$\Rightarrow C: \mathbf{r}(t) = \sqrt{3} \cos t \mathbf{i} - \sqrt{3} \sin t \mathbf{j} - \mathbf{k}, 0 \leq t \leq 2\pi$. $\Rightarrow$ parametrization

of C . $\frac{d\mathbf{r}}{dt} = d\mathbf{r} = -\sqrt{3} \sin t \mathbf{i} - \sqrt{3} \cos t \mathbf{j} + 0 \mathbf{k}$

$\Rightarrow \int_C F \cdot dr = \int_0^{2\pi} (\sqrt{3} \sin t - 3 \cos^2 t) dt = \sqrt{3} \int_0^{2\pi} \sin t dt - 3 \int_0^{2\pi} \cos^2 t dt$

$= -3 \left(\frac{\theta}{2} + \frac{\sin 2\theta}{2} \right) \Big|_0^{2\pi}$

$= -3 \left(\frac{2\pi}{2} \right) = -3\pi$